Posted on 15 Jul 2024 — The copyright holder is the author/funder. All rights reserved. No reuse without permission. — https://doi.org/10.22541/essoar.172108198.83331203/v1 — This is a preprint and has not been peer-reviewed. Data may be preliminary.

Reservoir Memory Complicates Water Management in the Upper Colorado River Basin

Benjamin L. Harding[1]

[1]Lynker Technologies, LLC

July 15, 2024

Abstract

Lake Powell regulates the erratic flows of the Upper Colorado River, storing water in wet periods for use during multi-decadal dry spells. Reservoirs also integrate consumptive use (CU) at or above the reservoir. In colloquial terms, a reservoir retains a memory of all previous inflows, depletions and outflows and it preserves this memory until it spills or empties completely. This article describes and demonstrates the nature of reservoir memory, explores the physical effects of persistence of reservoir memory, and illustrates what those physical effects mean for intra-state water management in the Upper Colorado River Basin. The period between when Lake Powell spills and subsequently empties will typically be several decades in length, so a small increase in Upper Basin CU will accumulate over that period and eventually lead to a large deficit in reservoir storage. This leads to two policy consequences: 1) a small annual exceedance of the amount of water legally available for CU in the Upper Basin can lead to a disruptive curtailment of water use under the terms of the Colorado River Compact, and 2) it is impossible to avoid or comply with a curtailment using involuntary cutbacks under a prior appropriation water rights system. A "grand bargain" that involves proportional allocation, or a cap on Upper Division CU in exchange for forbearance of curtailment by the Lower Basin in the event of shortfalls to the Lee Ferry flow obligation would eliminate this complication. This problem will also affect credit/debit-based entitlements.

Reservoir Memory Complicates Water Management in the Upper Colorado River Basin

Benjamin L. Harding[1]†

[1]Lynker Intel, Boulder, Colorado.

Corresponding author: Benjamin Harding (bharding@lynker.com)

†Lynker Intel, 5445 Conestoga Court, Suite 100, Boulder, Colorado 80301.

Key Points:

- Reservoirs integrate the effect of all upstream depletions after a spill until the reservoir reaches a lower operational lower limit.
- Accordingly, levels in Lake Powell reflect depletions from all water rights in the Upper Colorado River Basin.
- In the event of a curtailment of Upper Basin water use, effects of depletions by junior state rights could harm senior state water rights.

Abstract

Lake Powell regulates the erratic flows of the Upper Colorado River, storing water in wet periods for use during multi-decadal dry spells. Reservoirs also integrate consumptive use (CU) at or above the reservoir. In colloquial terms, a reservoir retains a memory of all previous inflows, depletions and outflows and it preserves this memory until it spills or empties completely. This article describes and demonstrates the nature of reservoir memory, explores the physical effects of persistence of reservoir memory, and illustrates what those physical effects mean for intra-state water management in the Upper Colorado River Basin. The period between when Lake Powell spills and subsequently empties will typically be several decades in length, so a small increase in Upper Basin CU will accumulate over that period and eventually lead to a large deficit in reservoir storage. This leads to two policy consequences: 1) a small annual exceedance of the amount of water legally available for CU in the Upper Basin can lead to a disruptive curtailment of water use under the terms of the Colorado River Compact, and 2) it is impossible to avoid or comply with a curtailment using involuntary cutbacks under a prior appropriation water rights system. A “grand bargain” that involves proportional allocation, or a cap on Upper Division CU in exchange for forbearance of curtailment by the Lower Basin in the event of shortfalls to the Lee Ferry flow obligation would eliminate this complication. This problem will also affect credit/debit-based entitlements.

Plain Language Summary

Lake Powell stores waters of the Colorado River and releases water to the Lower Basin. In the event that Lake Powell empties and flows to the Lower Basin subsequently fall below the levels mandated in the Colorado River Compact, a “compact call” could force curtailment of water use in the Upper Basin. Short of unpredictable outcomes from litigation by the Lower Basin, these curtailments would be coordinated by the Upper Colorado River Commission, even prospectively, and implemented by the states, all of which use prior appropriation (“first in time,

first in right") water rights systems. In those systems "junior" rights (those established later) are prohibited from diverting water when doing so would cause a shortfall to a "senior" right (one established earlier). Even if a junior right is diverting water only when it is "in priority", the consumptive use arising from that diversion will affect the levels in Lake Powell and could, decades later, precipitate, intensify or prolong a compact call, which would harm senior water rights. Without perfect foreknowledge of the natural flows of the Colorado River it is impossible to avoid this outcome.

1 Introduction

Lake Powell, impounded by Glen Canyon Dam, along with other reservoirs in the Upper Colorado River Basin, regulate the erratic flows of the Upper Colorado River, storing water in wet periods for use during multi-decadal dry spells. Reservoirs also integrate the effects of human activities: consumptive use at or above the reservoir, and releases from the reservoir. In colloquial terms, a reservoir retains a memory of all previous inflows, depletions and outflows and it preserves this memory until it spills or empties completely. This memory has effects on water management in the Upper Basin that have important policy consequences.

This article describes and demonstrates the nature of the integrating effect of a reservoir, termed "reservoir memory" here, characterizes the physical effects of long-term persistence of reservoir memory through a simulation based on a hypothetical but plausible physical and institutional scenario, and uses sensitivity analyses to illustrate what those physical effects mean for intra-state water management in the Upper Colorado River Basin.

This analysis shows that the interim period between the time Lake Powell spills and subsequently empties is typically decades in length; reservoir memory persists for that period—Upper Basin depletions accumulate over this interim period so a small increase in annual depletion will eventually lead to a large deficit in reservoir storage. This effect is straightforward, but its policy impact has not been broadly appreciated.

This article addresses two water management consequences that arise from those physical effects. One is that even a small exceedance above the amount of water legally available for consumptive use in the Upper Basin can lead to a disruptive curtailment of water use under the terms of the Colorado River Compact, which would injure senior water rights, and that it is impossible to avoid or comply with such a curtailment through involuntary cutbacks to water use without injury to senior water rights. To do so would require perfect foresight.

A "grand bargain" that involves a cap on depletions in the Upper Basin in exchange for forbearance of curtailment by the Lower Basin in the event of shortfalls to the Lee Ferry flow obligation could eliminate this consequence (CRGI, 2013; Kenney, 2019; Kuhn, 2012; Wheeler, 2019). Market-based voluntary prospective cutbacks or retrospective settlements, though complex, might be feasible management approaches under the existing legal framework.

2 The Upper Colorado River Basin Context

The Colorado River system is managed and operated in accordance with the "Law of the River", which consists of compacts, treaties, federal and state statutes, court decisions and decrees, contracts, and regulations. See Wilbur and Ely (1933), Wilbur and Ely (1948), Nathanson (1978), MacDonnell, et al. (1995), and Verburg (2010). A comprehensive and convenient compilation is at Weisheit (2010).

The Colorado River Compact (CRC, 1922) and the Upper Colorado River Basin Compact (UCRBC, 1948), the principal elements of the Law of the River, set constraints on consumptive use of water in the Upper Basin of the Colorado River. A water treaty with Mexico (1944) created a federal obligation to deliver water from the Colorado River system at the international border.

Article II of the CRC defines the Colorado River System and divides it into the Upper Basin and the Lower Basin. Lee Ferry, a point on the Colorado River main stem one mile below the mouth of the Paria River, divides the two basins. Article II also partitioned the states into the Upper Division (Colorado, New Mexico, Utah, and Wyoming) and the Lower Division (Arizona, California and Nevada). Arizona includes a small portion of the Upper Basin while New Mexico and Utah each include a small portion of the Lower Basin

Article III(a) of the CRC apportions to each basin 7.5 million acre-feet (maf) or 9.3 billion cubic meters (bcm) per year of consumptive use of the waters of the Colorado River System. Article III(b) grants an additional apportionment of 1 maf or 1.2 bcm per year of consumptive use to the Lower Basin.

The Mexican Water Treaty of 1944 established a federal obligation to deliver 1.5 maf (1.9 bcm) of water per year to Mexico. Article III(c) of the CRC sets out terms by which that treaty obligation would be shared between the Upper Division and the Lower Division--some portion of that federal delivery obligation may be the responsibility of the States of the Upper Division.

Article III(d) of the CRC sets out the terms of an obligation on the states of the Upper Division not to cause the 10-year cumulative flow at Lee Ferry to be depleted below 75 maf (92.5 bcm). This requirement is not without controversy (Kenney, 2012a; Kuhn, 2007; MacDonnell, 2021b). This flow obligation will likely be the most significant legal constraint on consumptive use in the Upper Basin. The framers of the Compact expected that annual natural flows at Lee Ferry would typically substantially exceed 16 maf (19.7 bcm), which would satisfy the 10-year cumulative flow obligation in Article III(d), allow the Upper Basin to consume 7.5 maf (9.3 bcm) under Article III(a) and to contribute one half of the treaty obligation to Mexico under Article III(c), and leave a surplus, but that expectation is now understood to have been optimistic (Kuhn and Fleck, 2019).

Article VIII of the CRC exempts Present Perfected Rights (PPRs) in the Upper Basin from the apportionment and obligations set out in Article III. PPRs are state water rights that were perfected before the execution of the Compact. Though there is controversy about their definition, most authorities define these as rights perfected before November 24th, 1922, the date the compact was signed. The effect of PPRs is to provide a reliable minimum supply of water in the Upper Division, but the tradeoff is that curtailment spells are extended.

Article III of the UCRBC apportions water from the Colorado River to the states in the Upper Basin. Arizona, a Lower Division state, was allocated a fixed amount of 50,000 acre-feet (af) or 62 million cubic meters (mcm). Water was apportioned among the Upper Division states according to percentage shares, reflecting the recognition, in the 26 years that had passed since the formulation of the CRC, that future Colorado River flows would likely be insufficient to supply the Upper Basin with its full Article III(a) apportionment (Kenney, 2012a).

The CRC apportioned water between the Upper and Lower Basins but established the Article III(d) non-depletion requirement as an obligation of the Upper Division. The UCRBC sets out rules for the curtailment of consumptive use in the states of the Upper Division. Water use in the Upper Basin in Arizona and water use in the Lower Basin in New Mexico and Utah are not subject to curtailment.

In general terms, if the flow at Lee Ferry falls below the obligations set out in Article III(c) and Article III(d)--however these may eventually be interpreted--consumptive use in the Upper Division states must be curtailed to the degree necessary to offset any flow shortfall, except that Present Perfected Rights are not subject to curtailment (Article IV, UCRBC, 1948). In this article the term "curtailment" is used to refer to the concept of a cutback required by Article III(d) of the CRC and Article IV of the UCRBC, and a particular such cutback. Curtailment is also used here to refer to the amount by which Upper Division consumptive use demand must be reduced in any year to comply with the Article III(d) obligation. Short of unpredictable outcomes from litigation by the Lower Basin, these curtailments would be coordinated by the Upper Colorado River Commission, even prospectively, and implemented by the states.

The principal reservoir in the Upper Basin is Lake Powell, with a current active capacity of 25 bcm (20.3 maf), impounded behind Glen Canyon Dam, about 29 kilometers above Lee Ferry. Construction of Glen Canyon Dam, closed in 1963, added another layer to the Law of the River in the form of operating rules: Long-range Operating Criteria (LROC, Reclamation, 1970) formulated under the Colorado River Basin Project Act (1968), as subsequently updated (Verburg, 2010). The provision in the LROC for a "minimum objective release" is most relevant for this study. The minimum objective release ("MOR") was set at 10.15 bcm (8.23 maf), consisting of the annualized value of the Article III(d) obligation (9.3 bcm (7.5 maf)), plus one-half of the federal obligation to Mexico (0.9 bcm (0.75 maf)), less the average annual flow of the Paria River (25 mcm (0.02 maf)) that enters the Colorado River between Glen Canyon Dam and the compact point at Lee Ferry.

In 2007, motivated by several years of drought and declining storage in the major reservoirs on the River, the Department of Interior established the Colorado River Interim Guidelines for Lower Basin Shortages and the Coordinated Operations for Lake Powell and Lake Mead (Interim Guidelines, Department of the Interior, 2007). Among other things, the Interim Guidelines provide for coordinated operations at Lake Powell and Lake Mead, based on levels in the two reservoirs, intended to "minimize shortages in the Lower Basin and avoid the risk of curtailments in the Upper Basin". The Interim Guidelines established four operating tiers in Lake Powell and three operating tiers in Lake Mead. When Lake Powell is in the lower three tiers, annual releases can range from 8.6 bcm (7.0 maf) to 11.7 bcm (9.5 maf), depending on the state of Lake Mead. When Lake Powell is in the upper, "Equalization Tier" releases greater than 11.7 bcm could occur to equalize storage in the two reservoirs or to avoid spills. The Interim Guidelines expire in 2026 and negotiations are underway to formulate their replacement. If these negotiations are unsuccessful operations may subsequently be controlled again by the LROC (Kuhn, 2024).

What might be the future interpretation of each of the elements of the Law of the River, and how they may be implemented, is broadly contentious and their precise nature will only be resolved by negotiation or prolonged litigation, or both (Robison & Kenney, 2012a; MacDonnell, 2021b).

3 The Intra-State Context—Water Rights Administration

All of the states of the Upper Division use prior appropriation ("first in time, first in right") water rights systems, established either constitutionally or statutorily. Administrative practices have been established by statute--it is the job of the state administrator to ensure that diverters do not take water out of priority.

The "linchpin" of the prior appropriation system is the notion of priority—the person who makes the earliest appropriation on a stream has the best right to the use of the water of the stream, and persons with later appropriations cannot impair use under that first appropriation. (Getches, 1984). More generally, a later appropriation, referred to as "junior", cannot cause injury to an earlier appropriation, referred to as "senior". The principle of non-injury rises from common law and was incorporated into the prior appropriation doctrine as the latter evolved. (Hobbs, 1997; MacDonnell, 2015).

Here, where I focus on compact curtailments, it is useful to categorize state water rights as "curtailment-senior", those that are entitled to continue to use water during a particular curtailment, and "curtailment-junior", those that will be cut back to meet the compact curtailment, while PPRs will never be cut back.

4 Reservoir Memory

This article is intended to raise awareness of how the Upper Division's Article III(d) non-depletion obligation and the integrating effect, "reservoir memory", of storage in Lake Powell combine to complicate water rights management and administration in the Upper Division states, all of which use a variant of the prior appropriation doctrine. Figure 1 is an abstract illustration of reservoir memory comparing a baseline case with a perturbed case where additional depletions are assumed to occur upstream of the reservoir.

Figure 1 presents three events that illustrate how the effect of additional depletions would persist in reservoir storage, accumulate over time and be transferred to outflow when the reservoir spills or empties. In Figure 1, the baseline level of flow or storage is shown by the top of the column, the effect of the additional depletion is shown by the hatched portion and the resulting storage or flow is shown by the solid portion. The baseline flow (panel a) is further depleted (panel b) in year three (two units), year eight, (two units), year 10 (one unit) and year 20 (three units). In Panel c) the effect of a depletion persists in the reservoir until it spills (years 6, 22 and 23) or empties (year 14). In Panel d) all or part of the effect of additional depletion is eliminated from storage and transferred to outflow in years 6, 14, 22 and 23.

In the first event, an additional depletion of two units in year three reduces storage compared to baseline conditions by the same amount, and that effect persists until the reservoir spills in year six and transfers the effect to outflow. In the second event, an additional depletion of two units in year eight reduces storage compared to baseline conditions by the same amount. In year 10 an additional depletion of one unit further reduces storage, bringing the total accumulated reduction of storage compared to baseline conditions to three units, which persists until year 14, when the reservoir empties under baseline conditions. The effect of reduced storage then transfers to outflow. In the third event, an additional depletion of three units reduces reservoir storage in year 20. In year 22, the reservoir spilled two units under baseline conditions, so outflow is reduced by that amount. In year 23, when the reservoir spilled three units under baseline conditions, the remaining one unit of the storage effect is transferred to outflow.

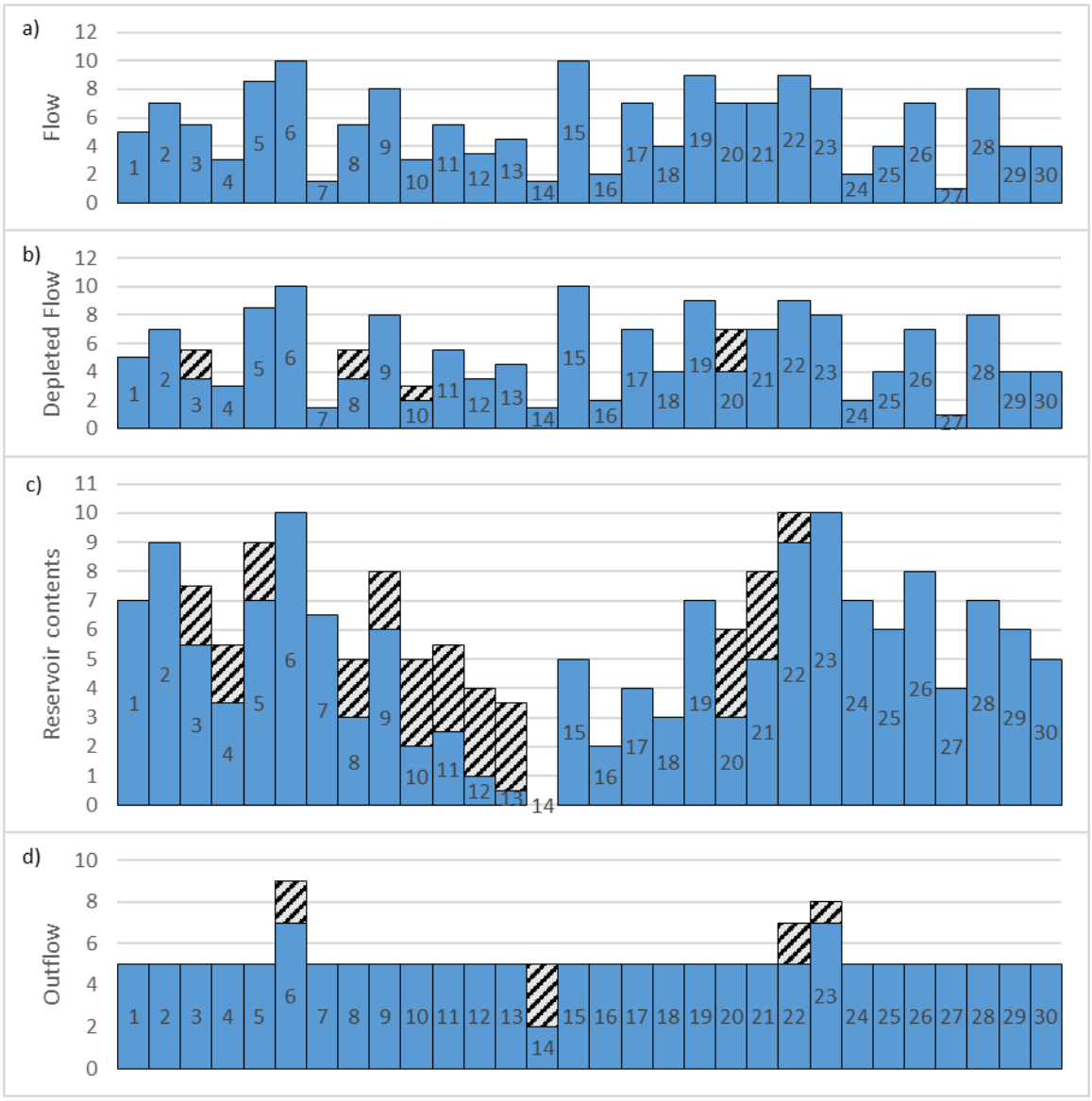


Figure 1. Reservoir Memory. This figure illustrates how reservoir memory affects storage and outflow, and how memory is transferred from the reservoir to the outflow when a reservoir spills or empties. In all panels, the top of each column represents baseline conditions; the hatched portion of a column is the effect of an additional upstream depletion, and the solid portion is the resulting storage or flow.

For clarity, Figure 1 represents a neat and simple set of conditions. The longest period over which depletions accumulate in the reservoir is six years in the second event. On the Colorado River, the results of this study suggest that the typical period from when Lake Powell spills to when it empties, extends over decades, so even small increases in annual Upper Division consumptive use (or small increases in annual releases at Glen Canyon Dam) will lead to significant reductions in storage. The remainder of this report provides quantitative illustrations, based on a plausible but hypothetical context, of the importance of reservoir memory.

5 About this Study

This analysis involves the interplay among elements of the Law of the River, Upper Division consumptive use of water, the physical behavior of Lake Powell and state water rights

systems. Figure 1 is an abstract illustration of the concept of reservoir memory. Illustrating the practical aspects requires a realistic context and this is established in this study with conditions and operations simulated against a plausible set of assumptions about inflow hydrology, water use and the law of the river. Sensitivity analyses using this simulation illustrate why reservoir memory will cause undue injury to compact-senior water rights in the event of a curtailment, as will, inevitably, prospective priority administration of state water rights to avoid a curtailment.

Assumptions about the Law of the River are the most problematic. The simple truth is that there can be little certainty about what exactly will become the Law of the River—that will probably be fought out over decade upon decade of litigation, by long and intense negotiation, or some combination of both (Kenney, 2012a; MacDonnell, 2021b; Robison & Kenney, 2012). The Article III(d) ten-year non-depletion obligation at Lee Ferry is the most significant legal assumption. Legal water availability in the Upper Division corresponds directly to this assumption. This is disputed due to the tension between it and the underlying principle of equity expressed in Article I and implied in the apportionment in Article III(a), and due to uncertainty about how this obligation should be interpreted if natural flow is reduced by climate change while depletions in the Upper Division remain unchanged (MacDonnell, 2021b; Robison and Kenney, 2012a). There is also substantial uncertainty and controversy about the apportionment of the Mexico delivery in Article III(c) (Kenney, 2012b; MacDonnell, 2021b).

Only slightly less uncertain are the true future hydrologic conditions under climate change. The reconstructed flows used here represent an assumption of stationarity wherein the mean and variability of future flows will be similar to the historical and prehistorical past. Climate change makes this assumption untenable for quantitative estimates of future water availability. Although it is possible that future long-term average Colorado River flows will be greater than the current historical average, such an increase is very unlikely (Harding, et al., 2012; Udall & Overpeck, 2017). However, the effect on water rights administration illustrated here arises from multi-decadal natural variability, and projections indicate that future variability will certainly differ but will likely resemble that evident in estimates of prehistoric flows (Miller et al., 2013).

Each of these assumptions will be disputed in one way or another by some stakeholder in the Colorado River Basin. However, these simplified and particular assumptions are adequate for the objectives of this analysis. While the magnitude and pattern of the effects quantified herein will change with different assumptions about inflow hydrology, Upper Division consumptive use, and the Law of the River, it is not apparent how these effects could vanish. Finally, it is important to understand that the objective of this study is not to make another set of predictions about the occurrence of critical periods and curtailments in the Upper Colorado River Basin, though those are offered here, but to use that information to illustrate the confounding nature of the interaction between the Law of the River, reservoir storage, and administration of intra-state water rights in the states of the Upper Division.

6 Approach

The simulation approach employed here (“Water Balance Model”) is an automation of the annual water balance method used in the U.S. Bureau of Reclamation’s 2007 Hydrologic Determination (2007HD, Reclamation, 2007a). The Hydrologic Determination is a yield study required by Congress before new contracts can be let for water supply from Navajo Reservoir. It is conducted from time to time by Reclamation and is the basis for approval by Congress of these

water contracts. The 2007HD is the most recent. The Water Balance Model was developed by reverse-engineering the results published in Reclamation (2007a). A similar approach, also based 2007HD, was used in the Colorado River Water Availability Study, Phase I (CWCB, 2012) and the Colorado River Water Bank Feasibility Study Phase 1 (MWH, 2012).

6.1 Water Balance Model

The Water Balance Model used in this work implemented equations 1 – 6 of 2007HD (Appendix A) to perform an annual water balance at Lee Ferry. The single inflow is the natural flow at Lee Ferry, the outflows are Upper Basin consumptive use (including Upper Basin PPRs and Arizona's Upper Basin CU), the Lee Ferry flow obligation, a portion of the Mexican Treaty delivery obligation, and shared CRSP reservoir evaporation. Table 1 lists the elements of the Water Balance Model. Specific elements of the model are described briefly below and in more detail in Appendix B.

The Water Balance Model was validated against 2007HD as described in Appendix C.

Table 1. Elements of the Water Balance Model

Element	Basis
Lees Ferry inflows	1,244-year reconstruction. (Meko, et al., 2007)
Upper Basin consumptive use demand	7.1 bcm (5.76 maf) per year. (Reclamation, 2007a)
Upper Division Present Perfected Right demand	2.8 bcm (2.267 maf) per year. (Kuhn, 2012)
Glen Canyon minimum objective release	10.15 bcm (8.23 maf) per year (Reclamation, 1970)
Lee Ferry composite flow obligation	101.8 bcm 82.5 maf per10 years
Mexican Treaty obligation	0.9 bcm (0.75 maf) per year (Upper Division share)
Upper Basin reservoir capacity	36.4 bcm (29.53 maf) (Reclamation, 2007a)
Reservoir evaporation	Reclamation, 2007a. Refer to Appendices A and B

6.2 Inflows

The simulation is forced with the 1244-year (762-2005) reconstruction of prehistoric natural flows at Lees Ferry averaging 18.1 bcm (14.7 maf) per year reported in Meko, et al. (2007). This is the reconstruction used in Appendix N of the *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead* (Reclamation, 2007b), the Colorado River Water Availability Study, Phase I (CWCB, 2012) and in the Colorado River Basin Water Supply and Demand Study (Reclamation, 2012). Its use here facilitates comparison with those earlier reports.

6.3 Upper Basin Consumptive Use Demand

Upper Basin consumptive use demand (UBCD) is a constant parameter representing an estimate of the total annual amount of consumptive use in the Upper Basin, including the fixed 62 mcm (50,000 af) of Upper Basin consumptive use apportioned to Arizona. Upper Basin consumptive use does not include shared CRSP evaporation but includes evaporation from non-CRSP reservoirs and Navajo Reservoir. In the baseline simulation, UBCD was set to 7.1 bcm or 5.76 maf, consistent with the formal yield determination of 2007HD, which represents the amount of water found to be physically and legally available for consumptive use in the Upper Basin under the assumptions adopted by 2007HD. Other simulations, used as sensitivity analyses, were conducted where UBCD was set to 6.1 bcm (4.97 maf), 7.09bcm (5.75 maf) and

7.12 bcm (5.77 maf). A fourth sensitivity analysis cut back UBCD when reservoir levels were low. The motivation for and implications of these sensitivity analyses are described in the discussion.

6.4 Reservoir Capacity

All major reservoirs above Lee Ferry are assumed to be available to maintain flows past Lee Ferry, and are simulated as a single, lumped reservoir (Reclamation, 2007a, Appendix A). Reclamation (2007a) provides estimates of lumped reservoir active and live capacity in 2007 and in 2060, based on estimates of sediment accumulation. In this analysis, lumped reservoir capacity was assumed to be 36.4 bcm (29.53 maf), representing active capacity (capacity above the power pools) available in 2060 considering estimated sedimentation. See Appendices A and B for more detail.

6.5 Simulated Operation Assumptions

Table 2 shows the conceptual priority of annual outflows as adopted in the analysis (1 being the highest priority) (MacDonnell, 2021a; Reclamation, 2007a).

Table 2. Priority of Outflows

Priority	Outflow
1	Reservoir evaporation,
2	Upper Division consumptive use under Present Perfected Rights,
3	The Upper Division Article III(c) portion of the Mexican Treaty Water delivery obligation,
4	The Article III(d) 10-year Lee Ferry non-depletion obligation,
5	Upper Basin consumptive use under non-PPRs,
6	Glen Canyon minimum objective release, and
7	Reservoir storage.

2007HD adopted the 10.18 bcm (8.25 maf) annual minimum objective release from the LROC, and I have done the same for this study, with an adjustment for compatibility with the inflow data set (see Appendix B).

As a simplifying assumption, 2007HD assumed that by meeting the MOR each year the Article III(d) ten-year flow obligation would be satisfied. This assumption was valid because in the 1906-2000 study period of 2007HD no shortfalls to the MOR occurred, but might not be appropriate for the longer study period simulated here, so the Water Balance Model included processes to simulate compliance with a 10-year cumulative obligation of 101.8 bcm (82.5 maf) at Lee Ferry, equivalent to a volume of 101.5 bcm (82.3 maf) at Glen Canyon Dam.

The model satisfies all Upper Basin consumptive use within the available physical and legal supply, regardless of the state of the reservoirs. While this assumption is not realistic—state water managers would certainly take actions to avoid a curtailment when reservoir storage dropped to alarming levels—it is tractable and meets the objective of this study.

Upper Division depletions are only curtailed, and resulting shortages reported, when necessary to satisfy a shortfall to the composite ten-year Lee Ferry flow obligation. When curtailments do occur, the Water Balance Model constrains cutbacks to protect consumptive use under PPRs in the Upper Division, and consumptive use in Arizona.

7 Results

This section reports the results of the baseline water balance simulations. Estimates of curtailment are reported for the Upper Division as a whole. See Harding, 2024.

7.1 Occurrence and Frequency of Curtailments and Curtailment Spells

Table 3 reports the statistics of curtailment spells in the baseline simulation. Over the course of the 1244-year simulation, 25 curtailment spells occurred, with lengths of 1-5 and 9 years. The average intensity (the average annual curtailment over the length of the spell) was 2.6 bcm (2.1 maf) and the weighted average length of the 25 spells was 2.2 years. The frequency (probability) of entering a curtailment spell and the return interval of a curtailment spell are 2% per year and 50 years, respectively. Generally, frequency tends to decrease as spell length increases, though only one spell occurred for each length of 4, 5, and 9 years. In all, curtailments occurred in 51 years, averaging 2.6 bcm (2.1 maf), so the frequency of a curtailment in any year is 4%.

Table 3. Curtailment Spells

Spell Length (years)	Events (#)	Average Intensity (bcm/maf)	Frequency (%/year)	Return Interval (years)
1	9	2.8/2.3	0.7	138
2	9	2.2/1.8	0.7	138
3	2	2.3/1.9	0.2	622
4	1	3.0/2.4	0.1	1244
5	1	3.0/2.4	0.1	1244
9	1	2.8/2.3	0.1	1244
All spells	23	2.6/2.1	1.8	51

Because this is a lumped model, it does not simulate local shortages on tributaries, where much of the consumptive use in the Upper Basin occurs. These local shortages are common in dry years and reduce the total Upper Basin CU, so these results overstate the frequency, duration and intensity of curtailments.

7.2 Interim Periods

The interval from one spill or curtailment to another curtailment is referred to here as an *interim period*. This is the amount of time over which depletions will accumulate in Lake Powell. Interim periods following spills range from 12 years to 73 years, with a mean of 35 years. Interim periods following curtailments are generally much shorter because of persistent low reservoir volumes, ranging from 1 year to 18 years and averaging 3 years. These short intervals allow less time for depletions to accumulate in storage.

8 Discussion

To illustrate how reservoir memory in Lake Powell confounds water rights administration in the Upper Division states we can look at a story, pulled from the 1244-year simulation. This story extends over 300 years and begins with the Upper Basin consuming 7.1 bcm (5.76 maf) of water over 236 years, free of a curtailment—the longest period in the baseline simulated record

without a curtailment. This period ends in 1120, after six straight years of spills at Lake Powell, which cleared any memory of prior depletions. Figure 2, which begins in 1115, shows two endings to this story.

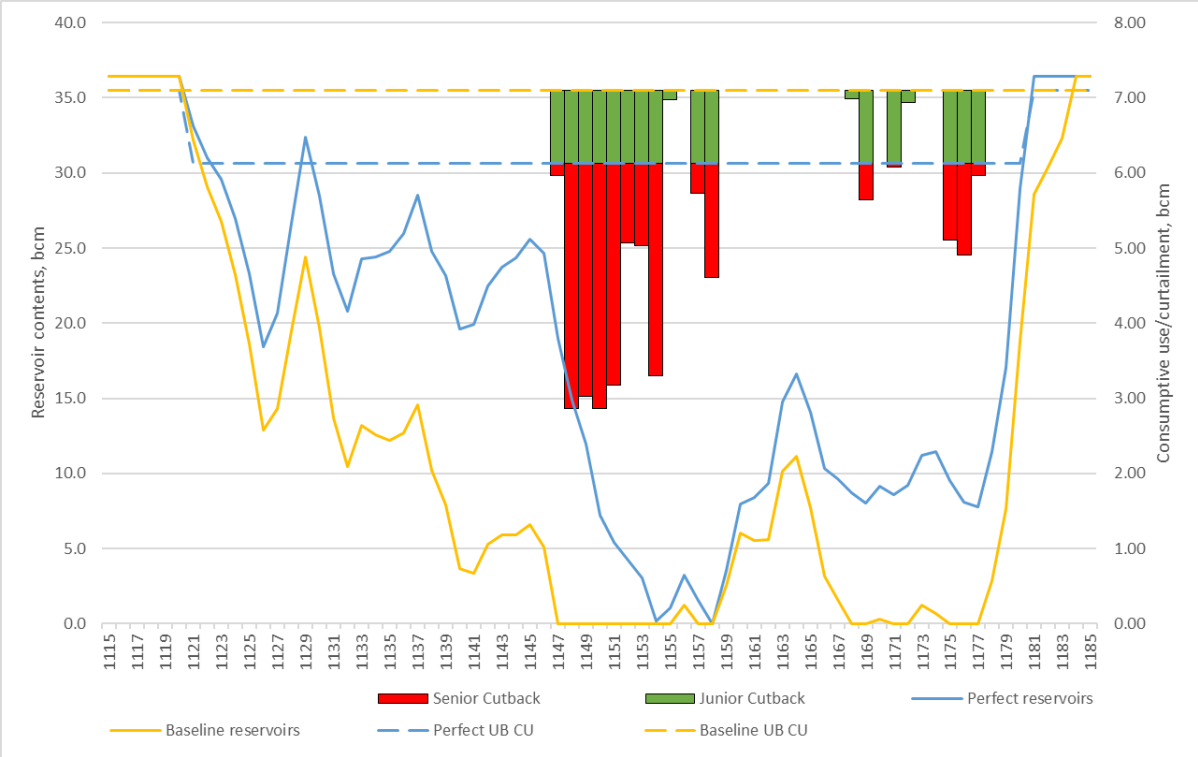


Figure 2. Inevitable harm to non-PPR curtailment-senior water rights. The dotted gold line shows the assumed steady Upper Basin consumptive use demand of 7.1 bcm (5.76 maf). The solid gold line represents the resulting reservoir contents, and the columns represent the cutbacks necessary to comply with the resulting curtailment, the red and green areas representing the cutbacks to the curtailment-junior and curtailment-senior uses, respectively. The solid and dotted blue lines represent the consequences of "perfect" administration under a constant cutback of Upper Basin consumptive use to 6.1 bcm (4.97 maf) from spill to spill; there are no curtailments in this case.

One ending goes like this: every October 1st, at the beginning of the water year, the state engineers of the Upper Division states gather and consult an oracle, asking what they should do in the next year. On October 1st, 1120, the oracle explains to the state engineers that it is not the next year that matters but the next 60 years. The oracle explains to them that their entitlement will only be 6.1 bcm (4.97 maf) beginning in 2021.

The state engineers return to their respective states and impose a water rights call such that collectively the Upper Division states will consume only 6.1 bcm (4.97 maf)—the "perfect call" for this critical period. This prospective perfect call could be seen as establishing the compact entitlement of the Upper Division under Articles III(c) and (d) and defining the non-PPR water rights that are curtailment-senior during this critical period and will be allowed to divert and consume water. Water use above the entitlement, under curtailment-junior rights, will be shut down. This saves 971 mcm (790,000 af) of water each year.

In Figure 2, the outcome of the perfect call is depicted by the blue lines. The dashed blue line represents the Upper Basin's entitlement during this critical period (6.1 bcm or 4.97 maf).

The solid blue line depicts the behavior of the upper basin reservoirs, primarily Lake Powell, which almost, but don't, empty on two occasions.

This ending to the story is impossible under our current state of knowledge: no such oracle has yet come forward, and human technology is understood never to be able to make perfect, or even useful predictions over such time frames.

Another possible ending to the story is this one: In this story there is no oracle—the state engineers must make their decisions year by year. After 1120 they wait and don't cut back water use as the reservoirs begin to drop, wishing to maximize overall water use in the Upper Division states, and believing there will always be a wet year to bail them out, as has been the case for the last 200-plus years. The outcome of "wishful thinking" is depicted by the gold lines in Figure 2: The level of consumptive use is shown by the dashed gold line, the resulting reservoir levels are shown by the solid gold line. The reservoir memory is apparent, as depletions accumulate and drain all storage completely for extended and multiple periods and result in large curtailments as shown by the columns reaching down from the dashed gold line. The portion of a curtailment that affects the curtailment-junior water rights is colored green in the columns; the portion of a curtailment affecting what would have been curtailment-senior water rights under a perfect call is shown in red. Wishful thinking leads to truly catastrophic curtailments of Upper Division water use; 18 years in all, totaling 38 bcm (31 maf); in 1148 and 1150 all non-PPR water rights in the Upper Division are curtailed. Curtailment-senior rights, which would have been able to continue diverting if the state engineers had exercised a perfect call, are cut back in 15 of those years, by a total volume of 23 bcm (18 maf). The state engineers do meet their objective of maximizing available water use: over the 71-year period an additional 32 bcm (26 maf) would be available for beneficial consumptive use (not including CRSP evaporation) in the Upper Division compared to the perfect call. About a third of this additional available water is from reduced evaporation due to the lower reservoir levels and the remaining two-thirds is due to reduced flows past Lee Ferry.

As is often said, "hindsight is 20/20", and in 1184 an observer of the wishful thinking case would be able to look back and know the Upper Division entitlement and what the perfect call should have been. From that they could establish which rights were curtailment-senior and which were curtailment-junior. If that observer is the owner of a curtailment-senior water right that had been curtailed, they would understand that they had suffered undue injury caused by water use under junior rights. The next thing they would do is call their lawyer.

This exact ending to wishful thinking is extremely unlikely—political leaders and the state engineers would at some point react to very low reservoir levels and order cutbacks of water use to try to avoid curtailments. Perhaps they would have in place some formalized "trigger" procedures to cut back water use when reservoir levels reached a certain point. Such approaches, as have been suggested in Colorado (Kuhn, 2012), might provide the means to avoid a curtailment, but if they use involuntary cutbacks, triggers can't avoid injury to seniors.

When the baseline case was simulated with the set of triggers shown in Table 4, the total number of cutbacks (triggered cutbacks and curtailments) increased from 18 to 40 and the total cutback volume increased by about 2% (11 bcm or 9 maf). More than half of this increase was due to several "false alarm" cutbacks during reservoir recovery that only served to increase spills. The remainder of the increase was due to evaporation from higher reservoir levels.

In addition to their inefficiency, the triggers did little to reduce undue injury to curtailment-senior rights. The number of and volume of cutbacks that affected curtailment-senior rights only decreased by about 10% (Harding, 2024).

Table 4. Trigger levels and cutbacks

Trigger Level (% of reservoir capacity)	Cutback to non-PPR rights (%)
> 33%	0
>25% - 33%	10%
>15% - 33%	20%
>10%-15%	40%
<= 10%	70%

The critical period shown in Figure 2, from 1121 to 1183, is the most severe dry spell in the 1244-year reconstruction, with a relatively short interim period of 27 years. I selected this period to illustrate the problem of injury to curtailment-senior rights for these reasons—the effects are obvious and dramatic. However more subtle exceedances of the UD entitlement can accumulate to have large effects over longer interim periods. The longest interim period in the baseline simulation was 73 years (1388 – 1460); increasing or decreasing UD depletions by 12 mcm (10,000 af) over that period increases or decreases the volume of curtailments in 1460-1461 by almost 500 mcm (400,000 af) (Harding, 2024).

Although it is tangential to this study, it is worth noting that releases from Lake Powell in excess of the MOR, which will happen under either the LROC or the Interim Guidelines, will be integrated into the reservoir state, accumulate as drawdowns, and may thereby increase the frequency, intensity, and duration of curtailments of consumptive use in the Upper Division states when releases to meet compact obligations cannot be made because of these accumulating drawdowns. Since 2007 more than 12.3 bcm (10 maf) of excess releases have accumulated as drawdowns in Lake Powell.

Attempts to avoid or comply with a curtailment using involuntary cutbacks of curtailment-senior rights will almost certainly be challenged, but voluntary, market-based cutbacks could be used. Reservoir levels could also inform an active market trading consumptive use forbearance or offsets among PPRs, non-PPR seniors and curtailment-juniors, a sort of futures market, which could serve to mitigate or even avoid a curtailment, or to settle residual inequities after curtailment occurs. These approaches are broadly analogous to augmentation plans for conjunctive use of groundwater (Getches, 1984). In the case of conjunctive use of groundwater, the time, place, and quantity of injury can be estimated, and thus planned for. The precision and accuracy of those estimates are relatively low, but the methods used to make them are broadly, if grudgingly, accepted. However, the time and extent of injury arising from a curtailment are highly variable and not known precisely until the curtailment occurs. If decades have passed since a spill, it is possible that the supply of offset or forbearance credits would be insufficient to settle all cumulative injury, in which case the price of offset supplies would exceed the means of some curtailment-juniors, or simply would not be available (in 1148-1151 and 1154 cutbacks to senior rights exceeded the amount of CU under PPRs). This suggests that without higher stakes many water users would simply not be able to settle their overdrafts. Disciplined enforcement of “water bankruptcy”, might lead to a workable approach to

reduce injury to curtailment-seniors. This is essentially what has happened in Colorado with the closure of wells with inadequate augmentation plans in the South Platte basin and to comply with compact terms in the Republican Basin and the Arkansas Basin (Associated Press, 2008; Colorado Mesa University, 2021). More sophisticated regulated market-based approaches such as some form of insurance or futures systems might be applicable (Broadbent, et al., 2017).

Eliminating the Lee Ferry Article III(d) obligation would eliminate the problem; this could come to pass through litigation or negotiation. The idea of a "grand bargain" on the Colorado River compares favorably with the alternative outcomes described above: unavoidable harm to non-PPR curtailment-seniors with consequential controversy and litigation, or the complexity and disruptive nature of market-based anticipatory or retrospective settlement approaches. In one concept of a grand bargain the Upper Division would agree to a cap on annual Upper Division consumptive use and the Lower Basin would, in exchange, agree not to call for water at Lee Ferry (CRGI, 2013; Kenney, 2019; Kuhn, 2012; Wheeler, 2019). A second type of grand bargain would be a proportional allocation system, with consumptive use allocations determined annually based on the actual annual water supply in the basin (Kuhn and Jacobs, 2022). Administering water rights to comply with a cap on depletions would easily be done on a long-term or year-to-year basis. Water management in the Upper Division states under these grand bargains would be far less complex and disruptive than the status quo, and the proportional approach offers the further advantage that water users within the Colorado River Basin would begin living within their means.

8 Acknowledgments

This work was supported by Lynker Intel, LLC. Dr. Graeme Aggett of Lynker gave much-appreciated encouragement. I am grateful to colleagues who reviewed drafts of this report, caught errors, and offered useful suggestions, particularly Robert Wigington, for his careful review of my writing about the Law of the River.

9 Open Research

The data supporting this analysis is available on the Open Science Framework (OSF) at DOI 10.17605/OSF.IO/4M9DN (Harding, 2024).

Appendix A. 2007 Hydrologic Determination Water Balance

The Hydrologic Determination water balance is an annual model of the water balance above Lee Ferry, including inflows, carryover reservoir contents, reservoir releases, reservoir evaporation, and Upper Basin consumptive use. The Hydrologic Determination water balance is subject to the physical constraints on reservoir storage in the Upper Basin, assumptions about the non-depletion flow obligation at Lee Ferry, and assumptions about apportionment of the federal delivery obligation to Mexico. It covers a study period of 1906 through 2000 and is based on a record of annual natural flows over that period. It established that the yield to the Upper Basin is 5.76 maf (7.1 bcm)

The Hydrologic Determination water balance model is a stepwise annual model involving five calculations (terms adopted from Reclamation [2007a]):

Water available to store = CR Natural Flow at Lee Ferry + Total Carryover storage
– Lower Basin Delivery – Upper Basin Use – Shared CRSP Evap; (A1)

Spill to LC = maximum(0, *Water available to Store – Adjusted Storage (2060)*); (A2)
Shortage = –minimum(0, *Water available to store*); (A3)
UC Basin Year-end Storage = Water available to store – Spill to LC + Shortage. (A4)
Total Carryover storage = UC Basin Year-end Storage (A5)

The independent terms are:

- *CR Natural Flow at Lee Ferry*—An annual time series of natural inflows at Lee Ferry from 1906 through 2000. Reclamation (2007a) describes this time series as being derived from the natural flow inputs to the Colorado River Simulation System (CRSS), with positive adjustments for flows during 1971-1980 to mitigate a reported error in estimates of Upper Basin consumptive use during that period. The inflows reported in Reclamation (2007a) do not match the inflows over the same period from the natural flow database currently used in CRSS (Reclamation, 2019).

- *Adjusted Storage (2060)*—A constant representing the total reservoir capacity available above Lee Ferry, assuming accumulated sediment as of 2060. If the power pools at Colorado River Storage Project (CRSP) units are assumed to be available for active storage this variable is set to 41,733 mcm (33,833,590 af) (live capacity), otherwise this variable is set to 36,425 mcm (29,530,030 af) (active capacity).

- Because the CRSP and non-CRSP reservoirs are lumped together they are assumed to be drawn down and filled in proportion.

- *Total Carryover storage*—I infer that at year one (1906 in Reclamation, 2007a) this is initialized to *Adjusted Storage (2060).* In each subsequent annual simulation *Total Carryover storage* is set equal to *UC Basin Year-end Storage* for the previous year (Equation 5).

- *Lower Basin Delivery*—This is a constant. In four simulations in Reclamation (2007a) it is set to 10.2 bcm (8.25 maf); for the other four it is set to 9.3 bcm (7.50 maf). The former value is used for the final estimate of water availability adopted by the 2007 Hydrologic Determination. The value of 9.3 bcm (7.50 maf) represents the minimum annual flow at Lee Ferry that would satisfy the ten-year flow obligation in Article III(d) of the CRC. The value of 10.2 bcm (8.25 maf) includes inflows from the Paria River and represents the minimum annual flow sufficient to satisfy the Article III(d) flow obligation plus one-half of the federal 1.9 bcm (1.5 maf) delivery obligation to Mexico under the Mexican Water Treaty (1944). If the states of the Upper Division must bear transit losses, as discussed earlier, the requirement at Lee Ferry could be more than 0.9 bcm (0.75 maf) while in times of "surplus" it could be less or zero.

- *Upper Basin Use*—This is a constant representing the nominal demand for consumptive use of water in the Upper Basin (including non-CRSP reservoir evaporation). (This is termed "Upper Basin consumptive use demand" in the water balance model.) An annual value of *Upper Basin Use* adjusted for shortages (curtailed use) is not calculated in the Hydrologic Determination water balance, but the total shortage volume and total shortage percentage (calculated against *Upper Basin Use*) for the study period and for 1953–1977 and 1931–1977 are reported. The Hydrologic Determination water balance does not simulate local shortages in the Upper Basin.

- *Shared CRSP Evap*—This is the evaporation from CRSP reservoirs that is shared among the Upper Division States. It is limited to evaporation from Lake Powell, Flaming Gorge Reservoir, and the Aspinall Unit and excludes evaporation from Navajo Reservoir which is

charged only to New Mexico because the supply from Navajo Reservoir is contracted for consumptive use only in that state and the reservoir is not currently available to meet Lee Ferry obligations. Intra-state reservoir evaporation is considered part of Upper Basin consumptive use.

Reclamation (2007a) documents two equations said to be used for calculating CRSP reservoir evaporation for each year, but values calculated using these equations do not match exactly the values reported in the water balance results. Regression analysis of alternative formulations for estimating evaporation identified Equation A6 as the formulation used in Run 2 of the 2007 Hydrologic Determination. That formulation has been used in the Water Balance Model. Equation A7 has not been verified.

Shared CRSP Evap = 0.020874 x mean(*Total Carryover storage*, *UC Basin Year-end Storage*) + 132877 af (A6)

Shared CRSP Evap = 0.0257 x mean(*Total Carryover storage*, *UC Basin Year-end Storage*) + 5017 af (A7)

The Hydrologic Determination reported eight runs covering a range of assumptions. Run 2 was used as the basis for this work. It assumed active reservoir capacity as of 2060 (36.43 bcm or 29.53 maf), a Lee Ferry flow of 10.18 bcm (8.25 maf) and Upper Division consumptive use of 7.14 bcm (5.79 maf. Upper Basin depletions do not include CRSP evaporation.

Appendix B. Water Balance Model

The Water Balance Model used in this work implemented equations 1 – 6 of the 2007 Hydrologic Determination (Reclamation, 2007a, described in Appendix A) to perform an annual water balance at Lee Ferry.

B.1 Inflows

The simulation is forced with the 1244-year (762-2005) reconstruction of prehistoric natural flows at Lees Ferry reported in Meko, et al. (2007) averaging 18.1 bcm (14.7 maf) per year. This is the reconstruction used in Appendix N of the *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead* (Reclamation, 2007b), the Colorado River Water Availability Study, Phase I (CWCB, 2012) and in the Colorado River Basin Water Supply and Demand Study (Reclamation, 2012). Its use here facilitates comparison with those earlier reports.

The reconstructed flows from Meko et al. (2007) are based on a functional relationship between chronologies of tree-ring indices and historical natural flows. Meko et al. (2007) state that the reconstruction was based on flows at Lee Ferry (the compact point), and the supplemental information provided with the journal article describes the dataset as such, but data downloaded from the treeflow.info web site (Rice et al., 2009; treeflow.info, 2018) includes metadata that indicate that the reconstruction is based on natural flows at Lees Ferry. The reconstructed flows in both datasets are identical when rounded to the same precision. However, the data in both datasets labeled as observed flows are inconsistent between the two datasets and both are inconsistent with flows at either Lee Ferry or Lees Ferry (two different but close by locations as discussed below) in the current natural flow dataset from the U.S. Bureau of Reclamation (2019).

These inconsistencies, though not large, introduce some uncertainty with respect to the basis for the reconstruction. Based on information from one of the authors of Meko et al. (2007), the reconstruction has been assumed to be based on the Lees Ferry gage (Lucas, 2019).

The Lees Ferry gage is located approximately one mile above the mouth of the Paria River and approximately two miles above the compact division point at Lee Ferry. Flows at the Lees Ferry gage are assumed to be equal to releases from Glen Canyon Dam. When Lees Ferry flows are used to force the Water Balance Model, the Lee Ferry flow obligation is reduced by 25 mcm (20,000 af) per year, the average natural flow from the Paria River at its mouth, so that compliance with the Lee Ferry flow obligation can be assessed against Glen Canyon releases.

B.2 Upper Basin Consumptive Use Estimates

Upper Basin consumptive use demand (UBCD) is a constant parameter representing an estimate of the total annual amount of consumptive use resulting from full supply of all water rights in the Upper Basin, including the 62 mcm (50,000 af) of annual Upper Basin consumptive use apportioned to Arizona. UBCD does not include shared CRSP evaporation, but that evaporative loss is an element in the water balance above Lee Ferry. In the baseline simulation, UBCD was set to 7.1 bcm (5.76 maf), consistent with the formal yield determination of 2007HD. Note that none of the runs published in 2007HD used exactly this value; the closest value was 7.14 bcm (5.79 maf) in Run 2. Other simulations, used as sensitivity analyses, were conducted where UBCD was set to 6.1 bcm (4.97 maf), 7.09bcm (5.75 maf) and 7.12 bcm (5.77 maf). A fourth sensitivity analysis cut back UBCD when reservoir levels were low.

B.3 Quantification of Present Perfected Rights

Article VIII of the CRC exempts Present Perfected Rights (PPRs) from the apportionment and obligations set out in Article III. Kuhn (2012) offers an estimate of 2.8 bcm (2.267 maf) for the amount of consumptive use associated with Upper Basin PPRs, based on work by the Upper Colorado River Commission. That value was used in this simulation. This quantification does not account for extensive tribal water rights that might be considered like PPRs without yet having been put to use and that, once adjudicated and their senior priorities asserted, could substantially affect curtailments based on water right priorities.

The Water Balance Model constrains curtailments to protect consumptive use under PPRs in the Upper Basin. The effect of PPRs is to provide a reliable minimum supply of water in the Upper Basin, but the tradeoff is that curtailment spells are extended.

B.4 Treatment of Arizona Consumptive Use

Because the 62 mcm (50,000 af) of consumptive use apportioned to Arizona by the UCRBC is not subject to curtailment, it is lumped with the quantification of PPRs described above.

B.5 Article III Mexican Treaty Delivery Obligation

Article III(c) of the CRC sets out rules by which the 1.9 bcm (1.5 maf) per year federal delivery obligation to Mexico should be apportioned between the Upper Division and Lower Division states. Interpretation of these provisions is controversial (Kenney, 2012a; MacDonnell, 2021b). The federal obligation is often assumed to be equally apportioned between the Upper and Lower Divisions, but under a more severe interpretation, the Upper Division states may bear

transit losses on its share of the delivery obligation and may thus owe more than 0.9 bcm (0.75 maf) at Lee Ferry in some years (Wigington, 2019). A conventional interpretation of Article III(c) is that the Mexican Treaty delivery is an annual obligation. This analysis assumes that the Upper Division states are obligated to deliver 0.9 bcm (0.75 maf) every year at Lee Ferry as their share of the federal delivery obligation to Mexico.

B.6 Article III(d) Lee Ferry Flow Obligation and the Mexican Treaty Delivery Obligation

Article III(d) of the CRC restricts the Upper Basin from depleting the flow passing Lee Ferry below an aggregate of 92.5 bcm (75 maf) in any running ten-year period. This requirement is not without controversy (Kenney, 2012a; Kuhn, 2007; MacDonnell, 2021b)

As a simplifying assumption, 2007HD treated the Article III(d) ten-year flow obligation on an annual basis (9.3 bcm (7.5 maf)) and lumped that obligation with an assumed quantification of the Upper Basin share of the Mexican delivery obligation of 0.9 bcm (0.75 maf) per year, to create a composite annual flow obligation of 10.2 bcm (8.25 maf). This composite flow obligation was used in four runs to assess compliance with the Compact and the Mexican Treaty on a year-by-year basis. (2007HD also included four analyses that used a composite annual Lee Ferry flow obligation of 9.3 bcm (7.50 maf), reflecting an assumption that the Upper Basin will make no contribution to the federal delivery obligation to Mexico. See Appendix A, Table A1.) This year-by-year approach does not guarantee compliance with the cumulative obligation of Article III(d) but in none of these runs did the flow at Lee Ferry over a ten-year period fall below the implied ten-year composite cumulative flow obligation of 102 bcm (82.5 maf), thereby maintaining compliance with the assumed cumulative flow obligation of Article III(d) and the assumed annual delivery obligation under the Mexican Treaty.

Although treating the Lee Ferry flow obligation on a year-by-year basis was sufficient for 2007HD, doing so will not assure compliance with the assumption of a ten-year cumulative flow obligation under more severe conditions. Accordingly, the Water Balance Model assesses compliance with Article III(d) and Article III(c) by comparing the ten-year running sum of Lee Ferry flows against an assumed ten-year composite Lee Ferry flow obligation of 102 bcm (82.5 maf). Using this ten-year composite obligation simplifies the simulation, but whether this assures compliance with Article III(c) depends on the relative priorities assumed for obligations under Article III(d) and Article III(c). As described earlier, the Water Balance Model assumes that the Mexican Treaty delivery obligation is satisfied ahead of the Article III(d) flow obligation. Consequently, the assumed Upper Basin Article III(c) obligation will be satisfied unless the simulated flow at Lee Ferry in any year falls below 0.9 bcm (0.75 maf), which does not happen in this simulation.

For consistency with Meko et al., 2007, the lumped Article III(d) and (c) ten-year flow obligation was quantified as 101.5 bcm (82.3 maf) at Lees Ferry, , accounting for 20,000 acre-feet/year of intervening inflows from the Paria River, and Upper Basin compliance with that obligation was evaluated using Glen Canyon releases (equivalent to Lees Ferry flows). These assumptions are consistent with most other analyses, including Reclamation (2007a), MWH (2013) and CWCB (2012).

Upper Basin depletions are only curtailed, and shortages reported, when necessary to satisfy a shortfall to the lumped ten-year Lee Ferry flow obligation.

B.7 Glen Canyon Dam Release

The same simplifying assumptions for Lee Ferry flow obligations adopted for this analysis and the 2007HD were made in establishing the minimum objective release (MOR) for Glen Canyon Dam as set out in the Criteria for Coordinated Long-Range Operation of Colorado River Reservoirs (Reclamation, 1970), now superseded until 2026 by Department of Interior (2007), "Interim Guidelines"; see also Colorado River Basin Project Act, section 602(a), 1968. The MOR is based on annualization of the Article III(d) obligation lumped with an assumed quantification of the Upper Basin's share of the federal Mexican delivery obligation for a total annual flow of 10.2 bcm (8.25 maf), as measured at Lee Ferry. Because Lee Ferry is located one mile below the mouth of the Paria River, which provides an average inflow of 25 mcm (20,000 af), the MOR is set to 10.15 bcm (8.23 maf). Glen Canyon Dam operations are now more complicated under the Interim Guidelines, but the results of this study are not affected by this simplified assumption.

B.8 Reservoir Capacity

All major reservoirs above Lee Ferry are assumed to be available to maintain flows past Lee Ferry, and are simulated as a single, lumped reservoir (Reclamation, 2007a, Appendix B). Reclamation (2007a) provides estimates of lumped reservoir capacity in 2007 and in 2060, based on estimates of sediment accumulation. For each time frame, Reclamation (2007a) provides estimates of storage capacity above the river outlets (live capacity) and above the minimum level at which hydropower generators can be operated (active capacity). In this analysis, lumped reservoir capacity was assumed to be 36.43 bcm (29.53 maf), representing active capacity available in 2060.

B.9 Evaporation

Under Article V(b)(1) of the 1948 UCRBC, evaporation from Lake Powell, Flaming Gorge Reservoir, and the Aspinall Unit is shared among the Upper Division states based on the states' percentage compact shares set out in Article III(a)(2) (shared CRSP evaporation). Evaporation at all other Upper Basin reservoirs is considered part of each state's share of Upper Basin consumptive use. (Evaporation from Navajo Reservoir, part of the CRSP, is chargeable to New Mexico based on the consumptive use of the Navajo supply within that state and is also included in Upper Basin consumptive use.) Reclamation (2007a) documents two equations used for calculating shared CRSP evaporation as a function of storage in the lumped reservoir, one for use when simulating active reservoir capacity and the second for use when simulating live reservoir capacity. These equations are documented in Appendix A (Equations A6 and A7). Because active reservoir capacity was used in this simulation, Equation 6 (Reclamation 2007a, Appendix A) was used to estimate reservoir evaporation.

Regression analysis determined that shared CRSP evaporation reported in Reclamation (2007a) was based on the average of reservoir contents at the beginning of the year and reservoir contents at the end of the year. Because year-end storage depends on the estimate of reservoir evaporation, an iterative approach is currently used for the annual simulation; referring to Appendix A, equations A1-A4 and A6 are iterated to convergence within 5 af to determine year-end storage. Carryover storage is then set (equation A5).

B.10 Curtailments

Curtailments are the amount by which UBCD must be reduced in any year to comply with the assumed composite flow obligation at Lee Ferry.

Appendix C. Water Balance Model Validation

The Water Balance Model was validated against Run 2, from Table A1 of 2007HD. Run 2 was used because it resulted in some shortages, which made comparison more straightforward. The validation run was conducted using the Lee Ferry natural flows from Reclamation (2007a), as these flows are different from those in Reclamation (2019) over the entire study period. The quantity of PPRs was set to zero, but this is a formality because the curtailments in the 2007 Hydrologic Determination did not reach the 2.8 bcm (2.267 maf) level described earlier. Because the water balance is simply an accounting, if the code is correct any error will be driven solely by the estimate of evaporation and will be very small.

The results of this simulation agreed very closely with the results for Run 2 published in Reclamation (2007a). Shortages, which are the variable of interest, and represent integration of conditions over extended periods, replicated values in Reclamation (2007a) within less than 0.001%.

This process did not test the validity of mechanisms in the Water Balance Model that were not exercised under the assumptions of Run 2. These mechanisms include protection of PPRs and conformance with the ten-year cumulative flow obligation at Lee Ferry. These mechanisms were validated by diagnosis and comparison to hand calculations.

References

Associated Press. 2008. Last-ditch effort to restart wells on Platte fails. Denver Post, May 3, 2008. https://www.denverpost.com/2008/05/03/last-ditch-effort-to-restart-wells-on-platte-fails/

Broadbent, C.D., Brookshire, D.S., Coursey, D. et al. 2017. Futures contracts in water leasing: an experimental analysis using basin characteristics of the Rio Grande, *NM. Environ Resource Econ* 68, 569–594 (2017). https://doi.org/10.1007/s10640-016-0032-4

Colorado Mesa University. 2021 Lessons Learned from Colorado Experiences with Interstate Compact Administration. https://www.coloradomesa.edu/water-center/compact-stories.html

Colorado River Basin Project Act. 82 stat 885. September 30, 1968. (See Nathanson, 1978.)

Colorado River Compact, November 24, 1922. *COLO. REV. STAT. § 37-61-101 (2012).* (See Nathanson, 1978.)

Colorado River Governance Initiative (CRGI). 2013. *The "Upper Basin Voluntary Demand Cap" as a means of mitigating legal uncertainty in the Colorado River Basin: modeling results*. April, 2013

Colorado River Governance Initiative (CRGI). 2010. *Rethinking the future of the Colorado River*, Draft Interim Report of the Colorado River Governance Initiative. December, 2010

Colorado Water Conservation Board (CWCB). 2012. *Colorado River water availability study*. Colorado Water Conservation Board. https://dnrweblink.state.co.us/cwcb/0/doc/158319/Electronic.aspx?searchid=78f0eafa-0b8f-4d8a-9ff3-faf67cc82f52.

Department of the Interior. 2007. Record of Decision Colorado River interim guidelines for lower basin shortages and the coordinated operations for Lake Powell and Lake Mead. U.S.

764 Department of Interior, December 2007.
765 https://www.usbr.gov/lc/region/programs/strategies/RecordofDecision.pdf
766 Getches, David. H. 1984. *Water law in a nutshell*. St. Paul MN, Thomson/Reuters.
767 Harding B.L., Wood A.W., Prairie J.R. 2012. The implications of climate change scenario
768 selection for future streamflow projection in the Upper Colorado River Basin. *Hydrol Earth*
769 *Syst Sci Discuss* 16: 3989–4007.
770 Harding, B.L. 2024.[Dataset] Reservoir Memory. https://doi.org/10.17605/OSF.IO/4M9DN
771 Hobbs, Gregory J. 1997. *Colorado water law: An historical overview*, 1 U. *Denv. Water L. Rev.* 1
772 Kenney, D. 2012a. Does the Upper Basin have a delivery obligation or an obligation not to
773 deplete the flow of the Colorado River at Lee Ferry? Western Water Policy Program,
774 Boulder Colorado, April 2012.
775 Kenney, D. 2012b. Respective obligations of the Upper and Lower Basins regarding the delivery
776 of water to Mexico: a review of key legal issues. Western Water Policy Program, Boulder
777 Colorado, March 2012.
778 Kenney, D. 2019. The case for the grand bargain. Colorado River District Annual Seminar.
779 Grand Junction, CO. September 18, 2019
780 Kuhn, E. 2012. Risk management strategies for the Upper Colorado River Basin. Colorado River
781 Water Conservation District. 2012. https://waterinfo.org/wp-
782 content/uploads/2019/01/Kuhn_on_Risk_Mgt_Strategies_of_the_UCRB.pdf
783 Kuhn, E. 2024. What Happens if there is no Agreement on Post-2026 Colorado River
784 Operations? https://www.inkstain.net/2024/04/what-happens-if-there-is-no-agreement-on-
785 post-2026-colorado-river-operations/
786 Kuhn, E. and Fleck, J. 2019. *Science be dammed: how ignoring inconvenient science drained the*
787 *Colorado River*. Tucson: University of Arizona Press. 288 pages.
788 Kuhn, E. and Jacobs, K. 2022. Robison, J.A. (2022). Alternative futures for the Colorado River system.
789 In *Cornerstone at the Confluence*: *Navigating the Colorado River Compact's Next Century*,
790 Robison, J.A. Ed., Tucson: University of Arizona Press, 2022.
791 https://doi.org/10.2307/j.ctv2z6qfp2.7
792 Lukas, Jeffery J., Personal communication, September 5, 2019.
793 MacDonnell, L., D.H. Getches, and W.C. Hugenberg Jr. 1995. The Law of the Colorado River:
794 Coping with Severe Sustained Drought. *Water Resources Bulletin*, 31:5(825-836), October.
795 MacDonnell, Lawrence. 2015. Prior Appropriation: A Reassessment, 18 U. *Denv. Water L. Rev.*
796 228 (2015)
797 MacDonnell, L. 2021a, Colorado River Basin. waters and water rights, Lexis-Nexus, CORB-1,
798 2021, Available at SSRN: https://ssrn.com/abstract=3780342
799 MacDonnell, L. 2021b. Sources of controversy in the law of the Colorado River: an Upper Basin
800 view. Available at SSRN: https://ssrn.com/abstract=3874212 or
801 http://dx.doi.org/10.2139/ssrn.3874212
802 Meko, D.M., C. A. Woodhouse, C. A. Baisan, T. Knight, J. J. Lukas, M. K. Hughes, & M. W.
803 Salzer. 2007. Medieval Drought in the Upper Colorado River Basin. *Geophysical Research*
804 *Letters* 34 (10). https://doi.org/10.1029/2007GL029988.
805 Mexican Water Treaty, 1944. *Treaty on utilization of the waters of the Colorado and Tijuana*
806 *Rivers and of the Rio Grande.* February 3, 1944, U.S. Government Printing Office, Treaty
807 Series 994. (See Nathanson, 1978.)
808 Miller, W. P., G.M. DeRosa, S. Gangopadhyay and J.B. Valdes. 2013. Predicting regime shifts in
809 flow of the Gunnison River under changing climate conditions. *Water Resources Research*,
810 v 49, 2966–2974, doi:10.1002/wrcr.20215.
811 MWH, 2012. *Colorado River Water Bank Feasibility Study – Phase 1, Final Draft Report.*
812 MWH Americas, Inc., Denver, Colorado. June, 2012.

Nathanson, M.N. 1978. *Updating the Hoover Dam Documents*. United States Department of the Interior Bureau of Reclamation. U.S. Government Printing Office, Denver, Colorado, 1980.

Reclamation, 1970. *Criteria for coordinated long-range operation of Colorado River reservoirs*. June, 1970.

Reclamation. 2007a. *Hydrologic determination: water availability from Navajo Reservoir and the Upper Colorado River Basin for use in New Mexico*. April, 2007.

Reclamation. 2007b. *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead;* Appendix N– Analysis of Hydrologic Variability Sensitivity. https://www.usbr.gov/lc/region/programs/strategies/FEIS/index.html. U.S. Department of the Interior, Bureau of Reclamation, Upper and Lower Colorado Regions. October, 2007.

Reclamation. 2012. *Colorado River Basin water supply and demand study, study report*. U.S. Department of Interior, Bureau of Reclamation. December, 2012.

Reclamation. 2019. *Natural flow and salt records*. https://www.usbr.gov/lc/region/g4000/NaturalFlow/NaturalFlows1906-2017_3.18.2019.xlsx Last accessed September 12, 2019. U.S. Department of the Interior, Bureau of Reclamation Lower Colorado Region, Boulder Canyon Operations Office.

Rice, J. L., C. A. Woodhouse, and J. J. Lukas. 2009. Science and decision making: water management and tree-ring data in the western united states. *JAWRA Journal of the American Water Resources Association* 45 (5): 1248–59. https://doi.org/10.1111/j.1752-1688.2009.00358.x.

Robison, J. A. & Kenney, D. S. 2012. Equity and the Colorado River Compact. *Environmental Law* 42, 1157–1209.

Treeflow.info. https://www.treeflow.info/ Data retrieved on November 20, 2018; last access on September 6, 2019.

Udall, B., and J. T. Overpeck. 2017: The twenty-first century Colorado River hot drought and implications for the future. *Water Resour. Res.*, 53, 2404–2418, doi:https://doi.org/10.1002/2016WR019638.

Upper Colorado River Basin Compact. 1948. October 11, 1948. (See Nathanson, 1978.)

Verburg, K. O. 2010. *The Colorado River documents 2008*. U.S. Department of the Interior, Bureau of Reclamation, Lower Colorado River Region. September 2010

Weisheit, J. 2019. *The Hoover Dam documents*. http://www.onthecolorado.com/articles.cfm?mode=detail&id=1292710182151 Accessed September 22, 2019.

Wheeler, K. 2019. The "Upper Basin Demand Cap": a new interbasin grand bargain? in *Charting a better course for the Colorado River: identifying the data and concepts to shape the interim guidelines renegotiation*. Getches-Wilkinson Center for Natural Resources, Energy, and the Environment. June, 2019.

Wigington, R. 2019. Personal communication. September 14, 2019.

Wilbur, R. L., Ely, N. 1933. *The Hoover Dam power and water contracts and related data*. United States Department of the Interior, United States Government Printing Office. Washington, D.C. 1933.

Wilbur, R. L., Ely, N. 1948. *The Hoover Dam documents*. 80th Congress 2nd Session, House Document No. 717. United States Government Printing Office. Washington, D.C. 1948.

Reservoir Memory Complicates Water Management in the Upper Colorado River Basin

Benjamin L. Harding[1]†

[1]Lynker Intel, Boulder, Colorado.

Corresponding author: Benjamin Harding (bharding@lynker.com)

†Lynker Intel, 5445 Conestoga Court, Suite 100, Boulder, Colorado 80301.

Key Points:

- Reservoirs integrate the effect of all upstream depletions after a spill until the reservoir reaches a lower operational lower limit.
- Accordingly, levels in Lake Powell reflect depletions from all water rights in the Upper Colorado River Basin.
- In the event of a curtailment of Upper Basin water use, effects of depletions by junior state rights could harm senior state water rights.

Abstract

Lake Powell regulates the erratic flows of the Upper Colorado River, storing water in wet periods for use during multi-decadal dry spells. Reservoirs also integrate consumptive use (CU) at or above the reservoir. In colloquial terms, a reservoir retains a memory of all previous inflows, depletions and outflows and it preserves this memory until it spills or empties completely. This article describes and demonstrates the nature of reservoir memory, explores the physical effects of persistence of reservoir memory, and illustrates what those physical effects mean for intra-state water management in the Upper Colorado River Basin. The period between when Lake Powell spills and subsequently empties will typically be several decades in length, so a small increase in Upper Basin CU will accumulate over that period and eventually lead to a large deficit in reservoir storage. This leads to two policy consequences: 1) a small annual exceedance of the amount of water legally available for CU in the Upper Basin can lead to a disruptive curtailment of water use under the terms of the Colorado River Compact, and 2) it is impossible to avoid or comply with a curtailment using involuntary cutbacks under a prior appropriation water rights system. A “grand bargain” that involves proportional allocation, or a cap on Upper Division CU in exchange for forbearance of curtailment by the Lower Basin in the event of shortfalls to the Lee Ferry flow obligation would eliminate this complication. This problem will also affect credit/debit-based entitlements.

Plain Language Summary

Lake Powell stores waters of the Colorado River and releases water to the Lower Basin. In the event that Lake Powell empties and flows to the Lower Basin subsequently fall below the levels mandated in the Colorado River Compact, a “compact call” could force curtailment of water use in the Upper Basin. Short of unpredictable outcomes from litigation by the Lower Basin, these curtailments would be coordinated by the Upper Colorado River Commission, even prospectively, and implemented by the states, all of which use prior appropriation (“first in time,

first in right") water rights systems. In those systems "junior" rights (those established later) are prohibited from diverting water when doing so would cause a shortfall to a "senior" right (one established earlier). Even if a junior right is diverting water only when it is "in priority", the consumptive use arising from that diversion will affect the levels in Lake Powell and could, decades later, precipitate, intensify or prolong a compact call, which would harm senior water rights. Without perfect foreknowledge of the natural flows of the Colorado River it is impossible to avoid this outcome.

1 Introduction

Lake Powell, impounded by Glen Canyon Dam, along with other reservoirs in the Upper Colorado River Basin, regulate the erratic flows of the Upper Colorado River, storing water in wet periods for use during multi-decadal dry spells. Reservoirs also integrate the effects of human activities: consumptive use at or above the reservoir, and releases from the reservoir. In colloquial terms, a reservoir retains a memory of all previous inflows, depletions and outflows and it preserves this memory until it spills or empties completely. This memory has effects on water management in the Upper Basin that have important policy consequences.

This article describes and demonstrates the nature of the integrating effect of a reservoir, termed "reservoir memory" here, characterizes the physical effects of long-term persistence of reservoir memory through a simulation based on a hypothetical but plausible physical and institutional scenario, and uses sensitivity analyses to illustrate what those physical effects mean for intra-state water management in the Upper Colorado River Basin.

This analysis shows that the interim period between the time Lake Powell spills and subsequently empties is typically decades in length; reservoir memory persists for that period—Upper Basin depletions accumulate over this interim period so a small increase in annual depletion will eventually lead to a large deficit in reservoir storage. This effect is straightforward, but its policy impact has not been broadly appreciated.

This article addresses two water management consequences that arise from those physical effects. One is that even a small exceedance above the amount of water legally available for consumptive use in the Upper Basin can lead to a disruptive curtailment of water use under the terms of the Colorado River Compact, which would injure senior water rights, and that it is impossible to avoid or comply with such a curtailment through involuntary cutbacks to water use without injury to senior water rights. To do so would require perfect foresight.

A "grand bargain" that involves a cap on depletions in the Upper Basin in exchange for forbearance of curtailment by the Lower Basin in the event of shortfalls to the Lee Ferry flow obligation could eliminate this consequence (CRGI, 2013; Kenney, 2019; Kuhn, 2012; Wheeler, 2019). Market-based voluntary prospective cutbacks or retrospective settlements, though complex, might be feasible management approaches under the existing legal framework.

2 The Upper Colorado River Basin Context

The Colorado River system is managed and operated in accordance with the "Law of the River", which consists of compacts, treaties, federal and state statutes, court decisions and decrees, contracts, and regulations. See Wilbur and Ely (1933), Wilbur and Ely (1948), Nathanson (1978), MacDonnell, et al. (1995), and Verburg (2010). A comprehensive and convenient compilation is at Weisheit (2010).

The Colorado River Compact (CRC, 1922) and the Upper Colorado River Basin Compact (UCRBC, 1948), the principal elements of the Law of the River, set constraints on consumptive use of water in the Upper Basin of the Colorado River. A water treaty with Mexico (1944) created a federal obligation to deliver water from the Colorado River system at the international border.

Article II of the CRC defines the Colorado River System and divides it into the Upper Basin and the Lower Basin. Lee Ferry, a point on the Colorado River main stem one mile below the mouth of the Paria River, divides the two basins. Article II also partitioned the states into the Upper Division (Colorado, New Mexico, Utah, and Wyoming) and the Lower Division (Arizona, California and Nevada). Arizona includes a small portion of the Upper Basin while New Mexico and Utah each include a small portion of the Lower Basin

Article III(a) of the CRC apportions to each basin 7.5 million acre-feet (maf) or 9.3 billion cubic meters (bcm) per year of consumptive use of the waters of the Colorado River System. Article III(b) grants an additional apportionment of 1 maf or 1.2 bcm per year of consumptive use to the Lower Basin.

The Mexican Water Treaty of 1944 established a federal obligation to deliver 1.5 maf (1.9 bcm) of water per year to Mexico. Article III(c) of the CRC sets out terms by which that treaty obligation would be shared between the Upper Division and the Lower Division--some portion of that federal delivery obligation may be the responsibility of the States of the Upper Division.

Article III(d) of the CRC sets out the terms of an obligation on the states of the Upper Division not to cause the 10-year cumulative flow at Lee Ferry to be depleted below 75 maf (92.5 bcm). This requirement is not without controversy (Kenney, 2012a; Kuhn, 2007; MacDonnell, 2021b). This flow obligation will likely be the most significant legal constraint on consumptive use in the Upper Basin. The framers of the Compact expected that annual natural flows at Lee Ferry would typically substantially exceed 16 maf (19.7 bcm), which would satisfy the 10-year cumulative flow obligation in Article III(d), allow the Upper Basin to consume 7.5 maf (9.3 bcm) under Article III(a) and to contribute one half of the treaty obligation to Mexico under Article III(c), and leave a surplus, but that expectation is now understood to have been optimistic (Kuhn and Fleck, 2019).

Article VIII of the CRC exempts Present Perfected Rights (PPRs) in the Upper Basin from the apportionment and obligations set out in Article III. PPRs are state water rights that were perfected before the execution of the Compact. Though there is controversy about their definition, most authorities define these as rights perfected before November 24th, 1922, the date the compact was signed. The effect of PPRs is to provide a reliable minimum supply of water in the Upper Division, but the tradeoff is that curtailment spells are extended.

Article III of the UCRBC apportions water from the Colorado River to the states in the Upper Basin. Arizona, a Lower Division state, was allocated a fixed amount of 50,000 acre-feet (af) or 62 million cubic meters (mcm). Water was apportioned among the Upper Division states according to percentage shares, reflecting the recognition, in the 26 years that had passed since the formulation of the CRC, that future Colorado River flows would likely be insufficient to supply the Upper Basin with its full Article III(a) apportionment (Kenney, 2012a).

The CRC apportioned water between the Upper and Lower Basins but established the Article III(d) non-depletion requirement as an obligation of the Upper Division. The UCRBC sets out rules for the curtailment of consumptive use in the states of the Upper Division. Water use in the Upper Basin in Arizona and water use in the Lower Basin in New Mexico and Utah are not subject to curtailment.

In general terms, if the flow at Lee Ferry falls below the obligations set out in Article III(c) and Article III(d)--however these may eventually be interpreted--consumptive use in the Upper Division states must be curtailed to the degree necessary to offset any flow shortfall, except that Present Perfected Rights are not subject to curtailment (Article IV, UCRBC, 1948). In this article the term "curtailment" is used to refer to the concept of a cutback required by Article III(d) of the CRC and Article IV of the UCRBC, and a particular such cutback. Curtailment is also used here to refer to the amount by which Upper Division consumptive use demand must be reduced in any year to comply with the Article III(d) obligation. Short of unpredictable outcomes from litigation by the Lower Basin, these curtailments would be coordinated by the Upper Colorado River Commission, even prospectively, and implemented by the states.

The principal reservoir in the Upper Basin is Lake Powell, with a current active capacity of 25 bcm (20.3 maf), impounded behind Glen Canyon Dam, about 29 kilometers above Lee Ferry. Construction of Glen Canyon Dam, closed in 1963, added another layer to the Law of the River in the form of operating rules: Long-range Operating Criteria (LROC, Reclamation, 1970) formulated under the Colorado River Basin Project Act (1968), as subsequently updated (Verburg, 2010). The provision in the LROC for a "minimum objective release" is most relevant for this study. The minimum objective release ("MOR") was set at 10.15 bcm (8.23 maf), consisting of the annualized value of the Article III(d) obligation (9.3 bcm (7.5 maf)), plus one-half of the federal obligation to Mexico (0.9 bcm (0.75 maf)), less the average annual flow of the Paria River (25 mcm (0.02 maf)) that enters the Colorado River between Glen Canyon Dam and the compact point at Lee Ferry.

In 2007, motivated by several years of drought and declining storage in the major reservoirs on the River, the Department of Interior established the Colorado River Interim Guidelines for Lower Basin Shortages and the Coordinated Operations for Lake Powell and Lake Mead (Interim Guidelines, Department of the Interior, 2007). Among other things, the Interim Guidelines provide for coordinated operations at Lake Powell and Lake Mead, based on levels in the two reservoirs, intended to "minimize shortages in the Lower Basin and avoid the risk of curtailments in the Upper Basin". The Interim Guidelines established four operating tiers in Lake Powell and three operating tiers in Lake Mead. When Lake Powell is in the lower three tiers, annual releases can range from 8.6 bcm (7.0 maf) to 11.7 bcm (9.5 maf), depending on the state of Lake Mead. When Lake Powell is in the upper, "Equalization Tier" releases greater than 11.7 bcm could occur to equalize storage in the two reservoirs or to avoid spills. The Interim Guidelines expire in 2026 and negotiations are underway to formulate their replacement. If these negotiations are unsuccessful operations may subsequently be controlled again by the LROC (Kuhn, 2024).

What might be the future interpretation of each of the elements of the Law of the River, and how they may be implemented, is broadly contentious and their precise nature will only be resolved by negotiation or prolonged litigation, or both (Robison & Kenney, 2012a; MacDonnell, 2021b).

3 The Intra-State Context—Water Rights Administration

All of the states of the Upper Division use prior appropriation ("first in time, first in right") water rights systems, established either constitutionally or statutorily. Administrative practices have been established by statute--it is the job of the state administrator to ensure that diverters do not take water out of priority.

The "linchpin" of the prior appropriation system is the notion of priority—the person who makes the earliest appropriation on a stream has the best right to the use of the water of the stream, and persons with later appropriations cannot impair use under that first appropriation. (Getches, 1984). More generally, a later appropriation, referred to as "junior", cannot cause injury to an earlier appropriation, referred to as "senior". The principle of non-injury rises from common law and was incorporated into the prior appropriation doctrine as the latter evolved. (Hobbs, 1997; MacDonnell, 2015).

Here, where I focus on compact curtailments, it is useful to categorize state water rights as "curtailment-senior", those that are entitled to continue to use water during a particular curtailment, and "curtailment-junior", those that will be cut back to meet the compact curtailment, while PPRs will never be cut back.

4 Reservoir Memory

This article is intended to raise awareness of how the Upper Division's Article III(d) non-depletion obligation and the integrating effect, "reservoir memory", of storage in Lake Powell combine to complicate water rights management and administration in the Upper Division states, all of which use a variant of the prior appropriation doctrine. Figure 1 is an abstract illustration of reservoir memory comparing a baseline case with a perturbed case where additional depletions are assumed to occur upstream of the reservoir.

Figure 1 presents three events that illustrate how the effect of additional depletions would persist in reservoir storage, accumulate over time and be transferred to outflow when the reservoir spills or empties. In Figure 1, the baseline level of flow or storage is shown by the top of the column, the effect of the additional depletion is shown by the hatched portion and the resulting storage or flow is shown by the solid portion. The baseline flow (panel a) is further depleted (panel b) in year three (two units), year eight, (two units), year 10 (one unit) and year 20 (three units). In Panel c) the effect of a depletion persists in the reservoir until it spills (years 6, 22 and 23) or empties (year 14). In Panel d) all or part of the effect of additional depletion is eliminated from storage and transferred to outflow in years 6, 14, 22 and 23.

In the first event, an additional depletion of two units in year three reduces storage compared to baseline conditions by the same amount, and that effect persists until the reservoir spills in year six and transfers the effect to outflow. In the second event, an additional depletion of two units in year eight reduces storage compared to baseline conditions by the same amount. In year 10 an additional depletion of one unit further reduces storage, bringing the total accumulated reduction of storage compared to baseline conditions to three units, which persists until year 14, when the reservoir empties under baseline conditions. The effect of reduced storage then transfers to outflow. In the third event, an additional depletion of three units reduces reservoir storage in year 20. In year 22, the reservoir spilled two units under baseline conditions, so outflow is reduced by that amount. In year 23, when the reservoir spilled three units under baseline conditions, the remaining one unit of the storage effect is transferred to outflow.

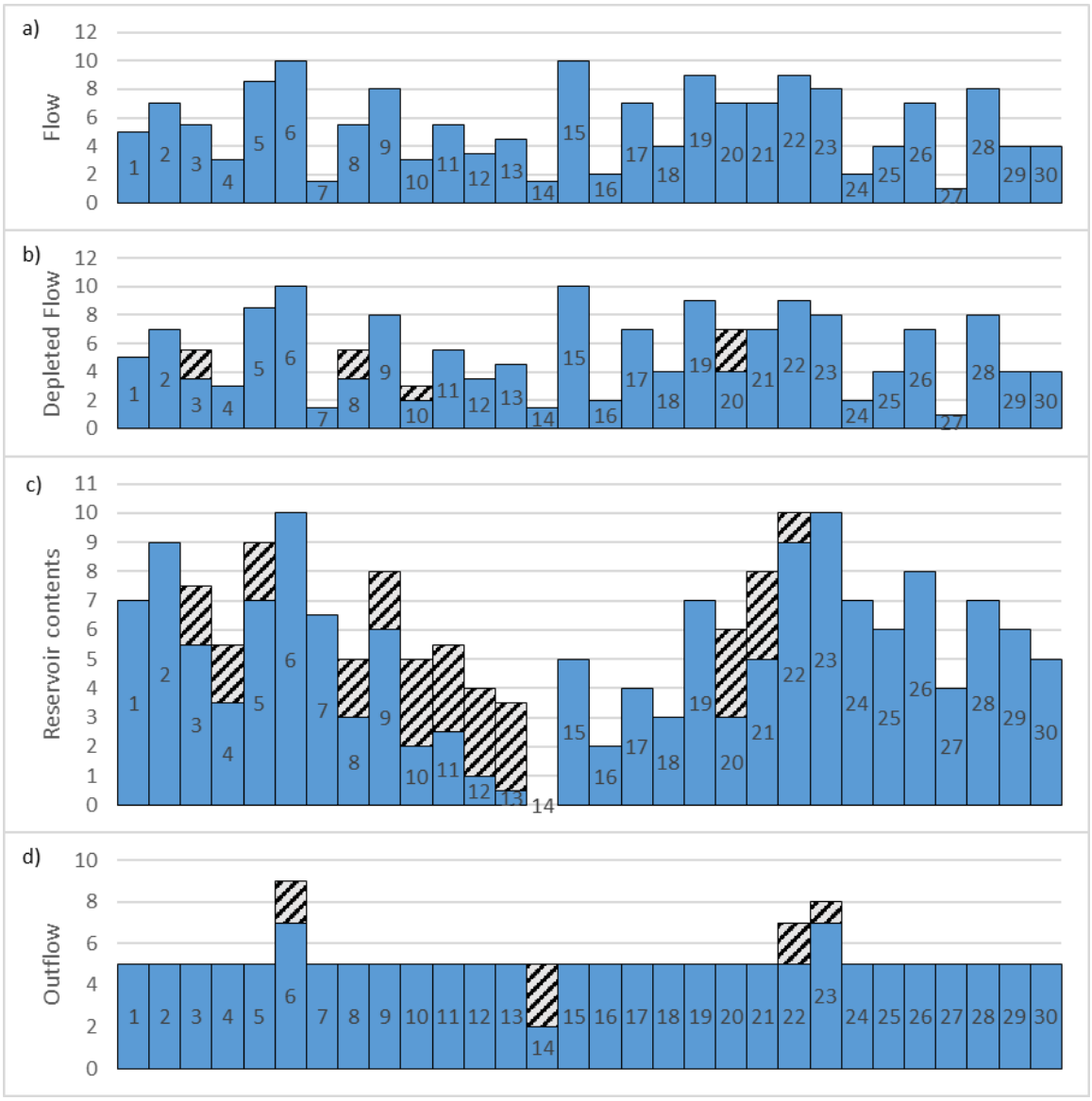


Figure 1. Reservoir Memory. This figure illustrates how reservoir memory affects storage and outflow, and how memory is transferred from the reservoir to the outflow when a reservoir spills or empties. In all panels, the top of each column represents baseline conditions; the hatched portion of a column is the effect of an additional upstream depletion, and the solid portion is the resulting storage or flow.

For clarity, Figure 1 represents a neat and simple set of conditions. The longest period over which depletions accumulate in the reservoir is six years in the second event. On the Colorado River, the results of this study suggest that the typical period from when Lake Powell spills to when it empties, extends over decades, so even small increases in annual Upper Division consumptive use (or small increases in annual releases at Glen Canyon Dam) will lead to significant reductions in storage. The remainder of this report provides quantitative illustrations, based on a plausible but hypothetical context, of the importance of reservoir memory.

5 About this Study

This analysis involves the interplay among elements of the Law of the River, Upper Division consumptive use of water, the physical behavior of Lake Powell and state water rights

systems. Figure 1 is an abstract illustration of the concept of reservoir memory. Illustrating the practical aspects requires a realistic context and this is established in this study with conditions and operations simulated against a plausible set of assumptions about inflow hydrology, water use and the law of the river. Sensitivity analyses using this simulation illustrate why reservoir memory will cause undue injury to compact-senior water rights in the event of a curtailment, as will, inevitably, prospective priority administration of state water rights to avoid a curtailment.

Assumptions about the Law of the River are the most problematic. The simple truth is that there can be little certainty about what exactly will become the Law of the River—that will probably be fought out over decade upon decade of litigation, by long and intense negotiation, or some combination of both (Kenney, 2012a; MacDonnell, 2021b; Robison & Kenney, 2012). The Article III(d) ten-year non-depletion obligation at Lee Ferry is the most significant legal assumption. Legal water availability in the Upper Division corresponds directly to this assumption. This is disputed due to the tension between it and the underlying principle of equity expressed in Article I and implied in the apportionment in Article III(a), and due to uncertainty about how this obligation should be interpreted if natural flow is reduced by climate change while depletions in the Upper Division remain unchanged (MacDonnell, 2021b; Robison and Kenney, 2012a). There is also substantial uncertainty and controversy about the apportionment of the Mexico delivery in Article III(c) (Kenney, 2012b; MacDonnell, 2021b).

Only slightly less uncertain are the true future hydrologic conditions under climate change. The reconstructed flows used here represent an assumption of stationarity wherein the mean and variability of future flows will be similar to the historical and prehistorical past. Climate change makes this assumption untenable for quantitative estimates of future water availability. Although it is possible that future long-term average Colorado River flows will be greater than the current historical average, such an increase is very unlikely (Harding, et al., 2012; Udall & Overpeck, 2017). However, the effect on water rights administration illustrated here arises from multi-decadal natural variability, and projections indicate that future variability will certainly differ but will likely resemble that evident in estimates of prehistoric flows (Miller et al., 2013).

Each of these assumptions will be disputed in one way or another by some stakeholder in the Colorado River Basin. However, these simplified and particular assumptions are adequate for the objectives of this analysis. While the magnitude and pattern of the effects quantified herein will change with different assumptions about inflow hydrology, Upper Division consumptive use, and the Law of the River, it is not apparent how these effects could vanish. Finally, it is important to understand that the objective of this study is not to make another set of predictions about the occurrence of critical periods and curtailments in the Upper Colorado River Basin, though those are offered here, but to use that information to illustrate the confounding nature of the interaction between the Law of the River, reservoir storage, and administration of intra-state water rights in the states of the Upper Division.

6 Approach

The simulation approach employed here (“Water Balance Model”) is an automation of the annual water balance method used in the U.S. Bureau of Reclamation’s 2007 Hydrologic Determination (2007HD, Reclamation, 2007a). The Hydrologic Determination is a yield study required by Congress before new contracts can be let for water supply from Navajo Reservoir. It is conducted from time to time by Reclamation and is the basis for approval by Congress of these

water contracts. The 2007HD is the most recent. The Water Balance Model was developed by reverse-engineering the results published in Reclamation (2007a). A similar approach, also based 2007HD, was used in the Colorado River Water Availability Study, Phase I (CWCB, 2012) and the Colorado River Water Bank Feasibility Study Phase 1 (MWH, 2012).

6.1 Water Balance Model

The Water Balance Model used in this work implemented equations 1 – 6 of 2007HD (Appendix A) to perform an annual water balance at Lee Ferry. The single inflow is the natural flow at Lee Ferry, the outflows are Upper Basin consumptive use (including Upper Basin PPRs and Arizona's Upper Basin CU), the Lee Ferry flow obligation, a portion of the Mexican Treaty delivery obligation, and shared CRSP reservoir evaporation. Table 1 lists the elements of the Water Balance Model. Specific elements of the model are described briefly below and in more detail in Appendix B.

The Water Balance Model was validated against 2007HD as described in Appendix C.

Table 1. Elements of the Water Balance Model

Element	Basis
Lees Ferry inflows	1,244-year reconstruction. (Meko, et al., 2007)
Upper Basin consumptive use demand	7.1 bcm (5.76 maf) per year. (Reclamation, 2007a)
Upper Division Present Perfected Right demand	2.8 bcm (2.267 maf) per year. (Kuhn, 2012)
Glen Canyon minimum objective release	10.15 bcm (8.23 maf) per year (Reclamation, 1970)
Lee Ferry composite flow obligation	101.8 bcm 82.5 maf per10 years
Mexican Treaty obligation	0.9 bcm (0.75 maf) per year (Upper Division share)
Upper Basin reservoir capacity	36.4 bcm (29.53 maf) (Reclamation, 2007a)
Reservoir evaporation	Reclamation, 2007a. Refer to Appendices A and B

6.2 Inflows

The simulation is forced with the 1244-year (762-2005) reconstruction of prehistoric natural flows at Lees Ferry averaging 18.1 bcm (14.7 maf) per year reported in Meko, et al. (2007). This is the reconstruction used in Appendix N of the *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead* (Reclamation, 2007b), the Colorado River Water Availability Study, Phase I (CWCB, 2012) and in the Colorado River Basin Water Supply and Demand Study (Reclamation, 2012). Its use here facilitates comparison with those earlier reports.

6.3 Upper Basin Consumptive Use Demand

Upper Basin consumptive use demand (UBCD) is a constant parameter representing an estimate of the total annual amount of consumptive use in the Upper Basin, including the fixed 62 mcm (50,000 af) of Upper Basin consumptive use apportioned to Arizona. Upper Basin consumptive use does not include shared CRSP evaporation but includes evaporation from non-CRSP reservoirs and Navajo Reservoir. In the baseline simulation, UBCD was set to 7.1 bcm or 5.76 maf, consistent with the formal yield determination of 2007HD, which represents the amount of water found to be physically and legally available for consumptive use in the Upper Basin under the assumptions adopted by 2007HD. Other simulations, used as sensitivity analyses, were conducted where UBCD was set to 6.1 bcm (4.97 maf), 7.09bcm (5.75 maf) and

7.12 bcm (5.77 maf). A fourth sensitivity analysis cut back UBCD when reservoir levels were low. The motivation for and implications of these sensitivity analyses are described in the discussion.

6.4 Reservoir Capacity

All major reservoirs above Lee Ferry are assumed to be available to maintain flows past Lee Ferry, and are simulated as a single, lumped reservoir (Reclamation, 2007a, Appendix A). Reclamation (2007a) provides estimates of lumped reservoir active and live capacity in 2007 and in 2060, based on estimates of sediment accumulation. In this analysis, lumped reservoir capacity was assumed to be 36.4 bcm (29.53 maf), representing active capacity (capacity above the power pools) available in 2060 considering estimated sedimentation. See Appendices A and B for more detail.

6.5 Simulated Operation Assumptions

Table 2 shows the conceptual priority of annual outflows as adopted in the analysis (1 being the highest priority) (MacDonnell, 2021a; Reclamation, 2007a).

Table 2. Priority of Outflows

Priority	Outflow
1	Reservoir evaporation,
2	Upper Division consumptive use under Present Perfected Rights,
3	The Upper Division Article III(c) portion of the Mexican Treaty Water delivery obligation,
4	The Article III(d) 10-year Lee Ferry non-depletion obligation,
5	Upper Basin consumptive use under non-PPRs,
6	Glen Canyon minimum objective release, and
7	Reservoir storage.

2007HD adopted the 10.18 bcm (8.25 maf) annual minimum objective release from the LROC, and I have done the same for this study, with an adjustment for compatibility with the inflow data set (see Appendix B).

As a simplifying assumption, 2007HD assumed that by meeting the MOR each year the Article III(d) ten-year flow obligation would be satisfied. This assumption was valid because in the 1906-2000 study period of 2007HD no shortfalls to the MOR occurred, but might not be appropriate for the longer study period simulated here, so the Water Balance Model included processes to simulate compliance with a 10-year cumulative obligation of 101.8 bcm (82.5 maf) at Lee Ferry, equivalent to a volume of 101.5 bcm (82.3 maf) at Glen Canyon Dam.

The model satisfies all Upper Basin consumptive use within the available physical and legal supply, regardless of the state of the reservoirs. While this assumption is not realistic—state water managers would certainly take actions to avoid a curtailment when reservoir storage dropped to alarming levels—it is tractable and meets the objective of this study.

Upper Division depletions are only curtailed, and resulting shortages reported, when necessary to satisfy a shortfall to the composite ten-year Lee Ferry flow obligation. When curtailments do occur, the Water Balance Model constrains cutbacks to protect consumptive use under PPRs in the Upper Division, and consumptive use in Arizona.

7 Results

This section reports the results of the baseline water balance simulations. Estimates of curtailment are reported for the Upper Division as a whole. See Harding, 2024.

7.1 Occurrence and Frequency of Curtailments and Curtailment Spells

Table 3 reports the statistics of curtailment spells in the baseline simulation. Over the course of the 1244-year simulation, 25 curtailment spells occurred, with lengths of 1-5 and 9 years. The average intensity (the average annual curtailment over the length of the spell) was 2.6 bcm (2.1 maf) and the weighted average length of the 25 spells was 2.2 years. The frequency (probability) of entering a curtailment spell and the return interval of a curtailment spell are 2% per year and 50 years, respectively. Generally, frequency tends to decrease as spell length increases, though only one spell occurred for each length of 4, 5, and 9 years. In all, curtailments occurred in 51 years, averaging 2.6 bcm (2.1 maf), so the frequency of a curtailment in any year is 4%.

Table 3. Curtailment Spells

Spell Length (years)	Events (#)	Average Intensity (bcm/maf)	Frequency (%/year)	Return Interval (years)
1	9	2.8/2.3	0.7	138
2	9	2.2/1.8	0.7	138
3	2	2.3/1.9	0.2	622
4	1	3.0/2.4	0.1	1244
5	1	3.0/2.4	0.1	1244
9	1	2.8/2.3	0.1	1244
All spells	23	2.6/2.1	1.8	51

Because this is a lumped model, it does not simulate local shortages on tributaries, where much of the consumptive use in the Upper Basin occurs. These local shortages are common in dry years and reduce the total Upper Basin CU, so these results overstate the frequency, duration and intensity of curtailments.

7.2 Interim Periods

The interval from one spill or curtailment to another curtailment is referred to here as an *interim period*. This is the amount of time over which depletions will accumulate in Lake Powell. Interim periods following spills range from 12 years to 73 years, with a mean of 35 years. Interim periods following curtailments are generally much shorter because of persistent low reservoir volumes, ranging from 1 year to 18 years and averaging 3 years. These short intervals allow less time for depletions to accumulate in storage.

8 Discussion

To illustrate how reservoir memory in Lake Powell confounds water rights administration in the Upper Division states we can look at a story, pulled from the 1244-year simulation. This story extends over 300 years and begins with the Upper Basin consuming 7.1 bcm (5.76 maf) of water over 236 years, free of a curtailment—the longest period in the baseline simulated record

without a curtailment. This period ends in 1120, after six straight years of spills at Lake Powell, which cleared any memory of prior depletions. Figure 2, which begins in 1115, shows two endings to this story.

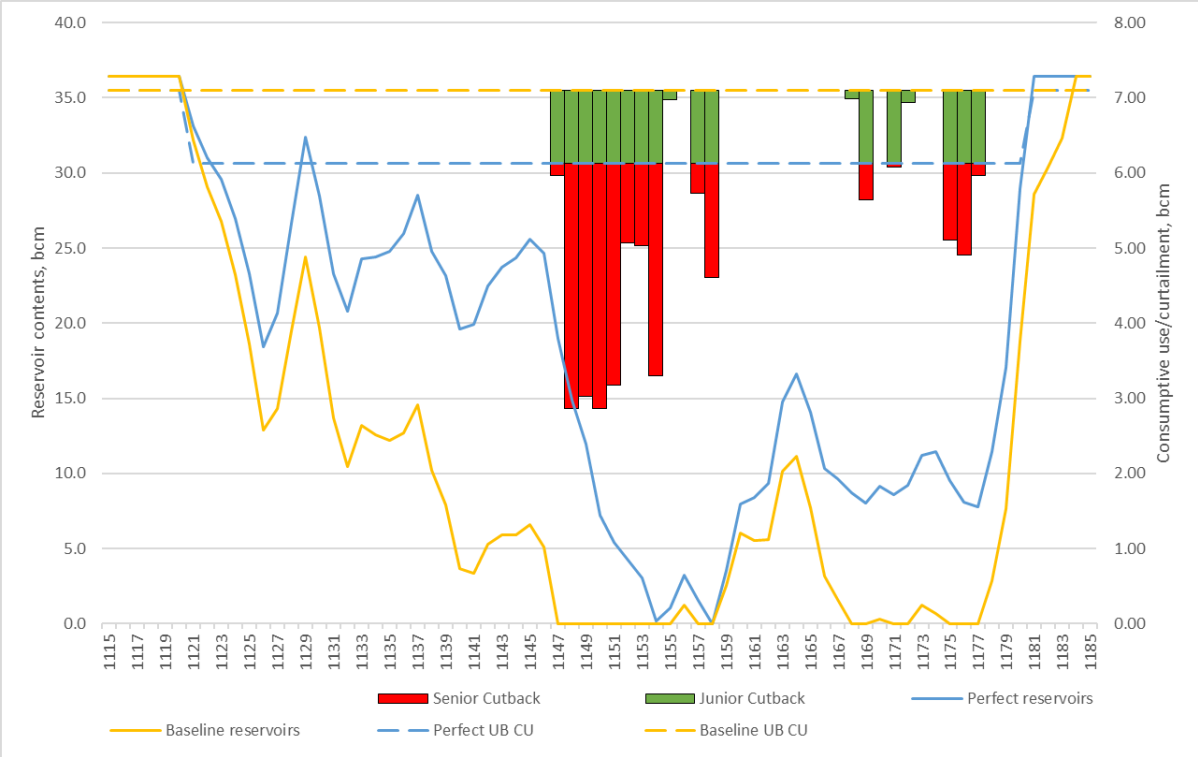


Figure 2. Inevitable harm to non-PPR curtailment-senior water rights. The dotted gold line shows the assumed steady Upper Basin consumptive use demand of 7.1 bcm (5.76 maf). The solid gold line represents the resulting reservoir contents, and the columns represent the cutbacks necessary to comply with the resulting curtailment, the red and green areas representing the cutbacks to the curtailment-junior and curtailment-senior uses, respectively. The solid and dotted blue lines represent the consequences of "perfect" administration under a constant cutback of Upper Basin consumptive use to 6.1 bcm (4.97 maf) from spill to spill; there are no curtailments in this case.

One ending goes like this: every October 1st, at the beginning of the water year, the state engineers of the Upper Division states gather and consult an oracle, asking what they should do in the next year. On October 1st, 1120, the oracle explains to the state engineers that it is not the next year that matters but the next 60 years. The oracle explains to them that their entitlement will only be 6.1 bcm (4.97 maf) beginning in 2021.

The state engineers return to their respective states and impose a water rights call such that collectively the Upper Division states will consume only 6.1 bcm (4.97 maf)—the "perfect call" for this critical period. This prospective perfect call could be seen as establishing the compact entitlement of the Upper Division under Articles III(c) and (d) and defining the non-PPR water rights that are curtailment-senior during this critical period and will be allowed to divert and consume water. Water use above the entitlement, under curtailment-junior rights, will be shut down. This saves 971 mcm (790,000 af) of water each year.

In Figure 2, the outcome of the perfect call is depicted by the blue lines. The dashed blue line represents the Upper Basin's entitlement during this critical period (6.1 bcm or 4.97 maf).

The solid blue line depicts the behavior of the upper basin reservoirs, primarily Lake Powell, which almost, but don't, empty on two occasions.

This ending to the story is impossible under our current state of knowledge: no such oracle has yet come forward, and human technology is understood never to be able to make perfect, or even useful predictions over such time frames.

Another possible ending to the story is this one: In this story there is no oracle—the state engineers must make their decisions year by year. After 1120 they wait and don't cut back water use as the reservoirs begin to drop, wishing to maximize overall water use in the Upper Division states, and believing there will always be a wet year to bail them out, as has been the case for the last 200-plus years. The outcome of "wishful thinking" is depicted by the gold lines in Figure 2: The level of consumptive use is shown by the dashed gold line, the resulting reservoir levels are shown by the solid gold line. The reservoir memory is apparent, as depletions accumulate and drain all storage completely for extended and multiple periods and result in large curtailments as shown by the columns reaching down from the dashed gold line. The portion of a curtailment that affects the curtailment-junior water rights is colored green in the columns; the portion of a curtailment affecting what would have been curtailment-senior water rights under a perfect call is shown in red. Wishful thinking leads to truly catastrophic curtailments of Upper Division water use; 18 years in all, totaling 38 bcm (31 maf); in 1148 and 1150 all non-PPR water rights in the Upper Division are curtailed. Curtailment-senior rights, which would have been able to continue diverting if the state engineers had exercised a perfect call, are cut back in 15 of those years, by a total volume of 23 bcm (18 maf). The state engineers do meet their objective of maximizing available water use: over the 71-year period an additional 32 bcm (26 maf) would be available for beneficial consumptive use (not including CRSP evaporation) in the Upper Division compared to the perfect call. About a third of this additional available water is from reduced evaporation due to the lower reservoir levels and the remaining two-thirds is due to reduced flows past Lee Ferry.

As is often said, "hindsight is 20/20", and in 1184 an observer of the wishful thinking case would be able to look back and know the Upper Division entitlement and what the perfect call should have been. From that they could establish which rights were curtailment-senior and which were curtailment-junior. If that observer is the owner of a curtailment-senior water right that had been curtailed, they would understand that they had suffered undue injury caused by water use under junior rights. The next thing they would do is call their lawyer.

This exact ending to wishful thinking is extremely unlikely—political leaders and the state engineers would at some point react to very low reservoir levels and order cutbacks of water use to try to avoid curtailments. Perhaps they would have in place some formalized "trigger" procedures to cut back water use when reservoir levels reached a certain point. Such approaches, as have been suggested in Colorado (Kuhn, 2012), might provide the means to avoid a curtailment, but if they use involuntary cutbacks, triggers can't avoid injury to seniors.

When the baseline case was simulated with the set of triggers shown in Table 4, the total number of cutbacks (triggered cutbacks and curtailments) increased from 18 to 40 and the total cutback volume increased by about 2% (11 bcm or 9 maf). More than half of this increase was due to several "false alarm" cutbacks during reservoir recovery that only served to increase spills. The remainder of the increase was due to evaporation from higher reservoir levels.

In addition to their inefficiency, the triggers did little to reduce undue injury to curtailment-senior rights. The number of and volume of cutbacks that affected curtailment-senior rights only decreased by about 10% (Harding, 2024).

Table 4. Trigger levels and cutbacks

Trigger Level (% of reservoir capacity)	Cutback to non-PPR rights (%)
> 33%	0
>25% - 33%	10%
>15% - 33%	20%
>10%-15%	40%
<= 10%	70%

The critical period shown in Figure 2, from 1121 to 1183, is the most severe dry spell in the 1244-year reconstruction, with a relatively short interim period of 27 years. I selected this period to illustrate the problem of injury to curtailment-senior rights for these reasons—the effects are obvious and dramatic. However more subtle exceedances of the UD entitlement can accumulate to have large effects over longer interim periods. The longest interim period in the baseline simulation was 73 years (1388 – 1460); increasing or decreasing UD depletions by 12 mcm (10,000 af) over that period increases or decreases the volume of curtailments in 1460-1461 by almost 500 mcm (400,000 af) (Harding, 2024).

Although it is tangential to this study, it is worth noting that releases from Lake Powell in excess of the MOR, which will happen under either the LROC or the Interim Guidelines, will be integrated into the reservoir state, accumulate as drawdowns, and may thereby increase the frequency, intensity, and duration of curtailments of consumptive use in the Upper Division states when releases to meet compact obligations cannot be made because of these accumulating drawdowns. Since 2007 more than 12.3 bcm (10 maf) of excess releases have accumulated as drawdowns in Lake Powell.

Attempts to avoid or comply with a curtailment using involuntary cutbacks of curtailment-senior rights will almost certainly be challenged, but voluntary, market-based cutbacks could be used. Reservoir levels could also inform an active market trading consumptive use forbearance or offsets among PPRs, non-PPR seniors and curtailment-juniors, a sort of futures market, which could serve to mitigate or even avoid a curtailment, or to settle residual inequities after curtailment occurs. These approaches are broadly analogous to augmentation plans for conjunctive use of groundwater (Getches, 1984). In the case of conjunctive use of groundwater, the time, place, and quantity of injury can be estimated, and thus planned for. The precision and accuracy of those estimates are relatively low, but the methods used to make them are broadly, if grudgingly, accepted. However, the time and extent of injury arising from a curtailment are highly variable and not known precisely until the curtailment occurs. If decades have passed since a spill, it is possible that the supply of offset or forbearance credits would be insufficient to settle all cumulative injury, in which case the price of offset supplies would exceed the means of some curtailment-juniors, or simply would not be available (in 1148-1151 and 1154 cutbacks to senior rights exceeded the amount of CU under PPRs). This suggests that without higher stakes many water users would simply not be able to settle their overdrafts. Disciplined enforcement of “water bankruptcy”, might lead to a workable approach to

reduce injury to curtailment-seniors. This is essentially what has happened in Colorado with the closure of wells with inadequate augmentation plans in the South Platte basin and to comply with compact terms in the Republican Basin and the Arkansas Basin (Associated Press, 2008; Colorado Mesa University, 2021). More sophisticated regulated market-based approaches such as some form of insurance or futures systems might be applicable (Broadbent, et al., 2017).

Eliminating the Lee Ferry Article III(d) obligation would eliminate the problem; this could come to pass through litigation or negotiation. The idea of a "grand bargain" on the Colorado River compares favorably with the alternative outcomes described above: unavoidable harm to non-PPR curtailment-seniors with consequential controversy and litigation, or the complexity and disruptive nature of market-based anticipatory or retrospective settlement approaches. In one concept of a grand bargain the Upper Division would agree to a cap on annual Upper Division consumptive use and the Lower Basin would, in exchange, agree not to call for water at Lee Ferry (CRGI, 2013; Kenney, 2019; Kuhn, 2012; Wheeler, 2019). A second type of grand bargain would be a proportional allocation system, with consumptive use allocations determined annually based on the actual annual water supply in the basin (Kuhn and Jacobs, 2022). Administering water rights to comply with a cap on depletions would easily be done on a long-term or year-to-year basis. Water management in the Upper Division states under these grand bargains would be far less complex and disruptive than the status quo, and the proportional approach offers the further advantage that water users within the Colorado River Basin would begin living within their means.

8 Acknowledgments

This work was supported by Lynker Intel, LLC. Dr. Graeme Aggett of Lynker gave much-appreciated encouragement. I am grateful to colleagues who reviewed drafts of this report, caught errors, and offered useful suggestions, particularly Robert Wigington, for his careful review of my writing about the Law of the River.

9 Open Research

The data supporting this analysis is available on the Open Science Framework (OSF) at DOI 10.17605/OSF.IO/4M9DN (Harding, 2024).

Appendix A. 2007 Hydrologic Determination Water Balance

The Hydrologic Determination water balance is an annual model of the water balance above Lee Ferry, including inflows, carryover reservoir contents, reservoir releases, reservoir evaporation, and Upper Basin consumptive use. The Hydrologic Determination water balance is subject to the physical constraints on reservoir storage in the Upper Basin, assumptions about the non-depletion flow obligation at Lee Ferry, and assumptions about apportionment of the federal delivery obligation to Mexico. It covers a study period of 1906 through 2000 and is based on a record of annual natural flows over that period. It established that the yield to the Upper Basin is 5.76 maf (7.1 bcm)

The Hydrologic Determination water balance model is a stepwise annual model involving five calculations (terms adopted from Reclamation [2007a]):

Water available to store = CR Natural Flow at Lee Ferry + Total Carryover storage
– Lower Basin Delivery – Upper Basin Use – Shared CRSP Evap; (A1)

Spill to LC = maximum(0, *Water available to Store – Adjusted Storage (2060)*); (A2)
Shortage = –minimum(0, *Water available to store*); (A3)
UC Basin Year-end Storage = Water available to store – Spill to LC + Shortage. (A4)
Total Carryover storage = UC Basin Year-end Storage (A5)

The independent terms are:

- *CR Natural Flow at Lee Ferry*—An annual time series of natural inflows at Lee Ferry from 1906 through 2000. Reclamation (2007a) describes this time series as being derived from the natural flow inputs to the Colorado River Simulation System (CRSS), with positive adjustments for flows during 1971-1980 to mitigate a reported error in estimates of Upper Basin consumptive use during that period. The inflows reported in Reclamation (2007a) do not match the inflows over the same period from the natural flow database currently used in CRSS (Reclamation, 2019).

- *Adjusted Storage (2060)*—A constant representing the total reservoir capacity available above Lee Ferry, assuming accumulated sediment as of 2060. If the power pools at Colorado River Storage Project (CRSP) units are assumed to be available for active storage this variable is set to 41,733 mcm (33,833,590 af) (live capacity), otherwise this variable is set to 36,425 mcm (29,530,030 af) (active capacity).

- Because the CRSP and non-CRSP reservoirs are lumped together they are assumed to be drawn down and filled in proportion.

- *Total Carryover storage*—I infer that at year one (1906 in Reclamation, 2007a) this is initialized to *Adjusted Storage (2060).* In each subsequent annual simulation *Total Carryover storage* is set equal to *UC Basin Year-end Storage* for the previous year (Equation 5).

- *Lower Basin Delivery*—This is a constant. In four simulations in Reclamation (2007a) it is set to 10.2 bcm (8.25 maf); for the other four it is set to 9.3 bcm (7.50 maf). The former value is used for the final estimate of water availability adopted by the 2007 Hydrologic Determination. The value of 9.3 bcm (7.50 maf) represents the minimum annual flow at Lee Ferry that would satisfy the ten-year flow obligation in Article III(d) of the CRC. The value of 10.2 bcm (8.25 maf) includes inflows from the Paria River and represents the minimum annual flow sufficient to satisfy the Article III(d) flow obligation plus one-half of the federal 1.9 bcm (1.5 maf) delivery obligation to Mexico under the Mexican Water Treaty (1944). If the states of the Upper Division must bear transit losses, as discussed earlier, the requirement at Lee Ferry could be more than 0.9 bcm (0.75 maf) while in times of "surplus" it could be less or zero.

- *Upper Basin Use*—This is a constant representing the nominal demand for consumptive use of water in the Upper Basin (including non-CRSP reservoir evaporation). (This is termed "Upper Basin consumptive use demand" in the water balance model.) An annual value of *Upper Basin Use* adjusted for shortages (curtailed use) is not calculated in the Hydrologic Determination water balance, but the total shortage volume and total shortage percentage (calculated against *Upper Basin Use*) for the study period and for 1953–1977 and 1931–1977 are reported. The Hydrologic Determination water balance does not simulate local shortages in the Upper Basin.

- *Shared CRSP Evap*—This is the evaporation from CRSP reservoirs that is shared among the Upper Division States. It is limited to evaporation from Lake Powell, Flaming Gorge Reservoir, and the Aspinall Unit and excludes evaporation from Navajo Reservoir which is

charged only to New Mexico because the supply from Navajo Reservoir is contracted for consumptive use only in that state and the reservoir is not currently available to meet Lee Ferry obligations. Intra-state reservoir evaporation is considered part of Upper Basin consumptive use.

Reclamation (2007a) documents two equations said to be used for calculating CRSP reservoir evaporation for each year, but values calculated using these equations do not match exactly the values reported in the water balance results. Regression analysis of alternative formulations for estimating evaporation identified Equation A6 as the formulation used in Run 2 of the 2007 Hydrologic Determination. That formulation has been used in the Water Balance Model. Equation A7 has not been verified.

Shared CRSP Evap = 0.020874 x mean(*Total Carryover storage*, *UC Basin Year-end Storage*) + 132877 af (A6)

Shared CRSP Evap = 0.0257 x mean(*Total Carryover storage*, *UC Basin Year-end Storage*) + 5017 af (A7)

The Hydrologic Determination reported eight runs covering a range of assumptions. Run 2 was used as the basis for this work. It assumed active reservoir capacity as of 2060 (36.43 bcm or 29.53 maf), a Lee Ferry flow of 10.18 bcm (8.25 maf) and Upper Division consumptive use of 7.14 bcm (5.79 maf. Upper Basin depletions do not include CRSP evaporation.

Appendix B. Water Balance Model

The Water Balance Model used in this work implemented equations 1 – 6 of the 2007 Hydrologic Determination (Reclamation, 2007a, described in Appendix A) to perform an annual water balance at Lee Ferry.

B.1 Inflows

The simulation is forced with the 1244-year (762-2005) reconstruction of prehistoric natural flows at Lees Ferry reported in Meko, et al. (2007) averaging 18.1 bcm (14.7 maf) per year. This is the reconstruction used in Appendix N of the *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead* (Reclamation, 2007b), the Colorado River Water Availability Study, Phase I (CWCB, 2012) and in the Colorado River Basin Water Supply and Demand Study (Reclamation, 2012). Its use here facilitates comparison with those earlier reports.

The reconstructed flows from Meko et al. (2007) are based on a functional relationship between chronologies of tree-ring indices and historical natural flows. Meko et al. (2007) state that the reconstruction was based on flows at Lee Ferry (the compact point), and the supplemental information provided with the journal article describes the dataset as such, but data downloaded from the treeflow.info web site (Rice et al., 2009; treeflow.info, 2018) includes metadata that indicate that the reconstruction is based on natural flows at Lees Ferry. The reconstructed flows in both datasets are identical when rounded to the same precision. However, the data in both datasets labeled as observed flows are inconsistent between the two datasets and both are inconsistent with flows at either Lee Ferry or Lees Ferry (two different but close by locations as discussed below) in the current natural flow dataset from the U.S. Bureau of Reclamation (2019).

These inconsistencies, though not large, introduce some uncertainty with respect to the basis for the reconstruction. Based on information from one of the authors of Meko et al. (2007), the reconstruction has been assumed to be based on the Lees Ferry gage (Lucas, 2019).

The Lees Ferry gage is located approximately one mile above the mouth of the Paria River and approximately two miles above the compact division point at Lee Ferry. Flows at the Lees Ferry gage are assumed to be equal to releases from Glen Canyon Dam. When Lees Ferry flows are used to force the Water Balance Model, the Lee Ferry flow obligation is reduced by 25 mcm (20,000 af) per year, the average natural flow from the Paria River at its mouth, so that compliance with the Lee Ferry flow obligation can be assessed against Glen Canyon releases.

B.2 Upper Basin Consumptive Use Estimates

Upper Basin consumptive use demand (UBCD) is a constant parameter representing an estimate of the total annual amount of consumptive use resulting from full supply of all water rights in the Upper Basin, including the 62 mcm (50,000 af) of annual Upper Basin consumptive use apportioned to Arizona. UBCD does not include shared CRSP evaporation, but that evaporative loss is an element in the water balance above Lee Ferry. In the baseline simulation, UBCD was set to 7.1 bcm (5.76 maf), consistent with the formal yield determination of 2007HD. Note that none of the runs published in 2007HD used exactly this value; the closest value was 7.14 bcm (5.79 maf) in Run 2. Other simulations, used as sensitivity analyses, were conducted where UBCD was set to 6.1 bcm (4.97 maf), 7.09bcm (5.75 maf) and 7.12 bcm (5.77 maf). A fourth sensitivity analysis cut back UBCD when reservoir levels were low.

B.3 Quantification of Present Perfected Rights

Article VIII of the CRC exempts Present Perfected Rights (PPRs) from the apportionment and obligations set out in Article III. Kuhn (2012) offers an estimate of 2.8 bcm (2.267 maf) for the amount of consumptive use associated with Upper Basin PPRs, based on work by the Upper Colorado River Commission. That value was used in this simulation. This quantification does not account for extensive tribal water rights that might be considered like PPRs without yet having been put to use and that, once adjudicated and their senior priorities asserted, could substantially affect curtailments based on water right priorities.

The Water Balance Model constrains curtailments to protect consumptive use under PPRs in the Upper Basin. The effect of PPRs is to provide a reliable minimum supply of water in the Upper Basin, but the tradeoff is that curtailment spells are extended.

B.4 Treatment of Arizona Consumptive Use

Because the 62 mcm (50,000 af) of consumptive use apportioned to Arizona by the UCRBC is not subject to curtailment, it is lumped with the quantification of PPRs described above.

B.5 Article III Mexican Treaty Delivery Obligation

Article III(c) of the CRC sets out rules by which the 1.9 bcm (1.5 maf) per year federal delivery obligation to Mexico should be apportioned between the Upper Division and Lower Division states. Interpretation of these provisions is controversial (Kenney, 2012a; MacDonnell, 2021b). The federal obligation is often assumed to be equally apportioned between the Upper and Lower Divisions, but under a more severe interpretation, the Upper Division states may bear

transit losses on its share of the delivery obligation and may thus owe more than 0.9 bcm (0.75 maf) at Lee Ferry in some years (Wigington, 2019). A conventional interpretation of Article III(c) is that the Mexican Treaty delivery is an annual obligation. This analysis assumes that the Upper Division states are obligated to deliver 0.9 bcm (0.75 maf) every year at Lee Ferry as their share of the federal delivery obligation to Mexico.

B.6 Article III(d) Lee Ferry Flow Obligation and the Mexican Treaty Delivery Obligation

Article III(d) of the CRC restricts the Upper Basin from depleting the flow passing Lee Ferry below an aggregate of 92.5 bcm (75 maf) in any running ten-year period. This requirement is not without controversy (Kenney, 2012a; Kuhn, 2007; MacDonnell, 2021b)

As a simplifying assumption, 2007HD treated the Article III(d) ten-year flow obligation on an annual basis (9.3 bcm (7.5 maf)) and lumped that obligation with an assumed quantification of the Upper Basin share of the Mexican delivery obligation of 0.9 bcm (0.75 maf) per year, to create a composite annual flow obligation of 10.2 bcm (8.25 maf). This composite flow obligation was used in four runs to assess compliance with the Compact and the Mexican Treaty on a year-by-year basis. (2007HD also included four analyses that used a composite annual Lee Ferry flow obligation of 9.3 bcm (7.50 maf), reflecting an assumption that the Upper Basin will make no contribution to the federal delivery obligation to Mexico. See Appendix A, Table A1.) This year-by-year approach does not guarantee compliance with the cumulative obligation of Article III(d) but in none of these runs did the flow at Lee Ferry over a ten-year period fall below the implied ten-year composite cumulative flow obligation of 102 bcm (82.5 maf), thereby maintaining compliance with the assumed cumulative flow obligation of Article III(d) and the assumed annual delivery obligation under the Mexican Treaty.

Although treating the Lee Ferry flow obligation on a year-by-year basis was sufficient for 2007HD, doing so will not assure compliance with the assumption of a ten-year cumulative flow obligation under more severe conditions. Accordingly, the Water Balance Model assesses compliance with Article III(d) and Article III(c) by comparing the ten-year running sum of Lee Ferry flows against an assumed ten-year composite Lee Ferry flow obligation of 102 bcm (82.5 maf). Using this ten-year composite obligation simplifies the simulation, but whether this assures compliance with Article III(c) depends on the relative priorities assumed for obligations under Article III(d) and Article III(c). As described earlier, the Water Balance Model assumes that the Mexican Treaty delivery obligation is satisfied ahead of the Article III(d) flow obligation. Consequently, the assumed Upper Basin Article III(c) obligation will be satisfied unless the simulated flow at Lee Ferry in any year falls below 0.9 bcm (0.75 maf), which does not happen in this simulation.

For consistency with Meko et al., 2007, the lumped Article III(d) and (c) ten-year flow obligation was quantified as 101.5 bcm (82.3 maf) at Lees Ferry, , accounting for 20,000 acre-feet/year of intervening inflows from the Paria River, and Upper Basin compliance with that obligation was evaluated using Glen Canyon releases (equivalent to Lees Ferry flows). These assumptions are consistent with most other analyses, including Reclamation (2007a), MWH (2013) and CWCB (2012).

Upper Basin depletions are only curtailed, and shortages reported, when necessary to satisfy a shortfall to the lumped ten-year Lee Ferry flow obligation.

B.7 Glen Canyon Dam Release

The same simplifying assumptions for Lee Ferry flow obligations adopted for this analysis and the 2007HD were made in establishing the minimum objective release (MOR) for Glen Canyon Dam as set out in the Criteria for Coordinated Long-Range Operation of Colorado River Reservoirs (Reclamation, 1970), now superseded until 2026 by Department of Interior (2007), "Interim Guidelines"; see also Colorado River Basin Project Act, section 602(a), 1968. The MOR is based on annualization of the Article III(d) obligation lumped with an assumed quantification of the Upper Basin's share of the federal Mexican delivery obligation for a total annual flow of 10.2 bcm (8.25 maf), as measured at Lee Ferry. Because Lee Ferry is located one mile below the mouth of the Paria River, which provides an average inflow of 25 mcm (20,000 af), the MOR is set to 10.15 bcm (8.23 maf). Glen Canyon Dam operations are now more complicated under the Interim Guidelines, but the results of this study are not affected by this simplified assumption.

B.8 Reservoir Capacity

All major reservoirs above Lee Ferry are assumed to be available to maintain flows past Lee Ferry, and are simulated as a single, lumped reservoir (Reclamation, 2007a, Appendix B). Reclamation (2007a) provides estimates of lumped reservoir capacity in 2007 and in 2060, based on estimates of sediment accumulation. For each time frame, Reclamation (2007a) provides estimates of storage capacity above the river outlets (live capacity) and above the minimum level at which hydropower generators can be operated (active capacity). In this analysis, lumped reservoir capacity was assumed to be 36.43 bcm (29.53 maf), representing active capacity available in 2060.

B.9 Evaporation

Under Article V(b)(1) of the 1948 UCRBC, evaporation from Lake Powell, Flaming Gorge Reservoir, and the Aspinall Unit is shared among the Upper Division states based on the states' percentage compact shares set out in Article III(a)(2) (shared CRSP evaporation). Evaporation at all other Upper Basin reservoirs is considered part of each state's share of Upper Basin consumptive use. (Evaporation from Navajo Reservoir, part of the CRSP, is chargeable to New Mexico based on the consumptive use of the Navajo supply within that state and is also included in Upper Basin consumptive use.) Reclamation (2007a) documents two equations used for calculating shared CRSP evaporation as a function of storage in the lumped reservoir, one for use when simulating active reservoir capacity and the second for use when simulating live reservoir capacity. These equations are documented in Appendix A (Equations A6 and A7). Because active reservoir capacity was used in this simulation, Equation 6 (Reclamation 2007a, Appendix A) was used to estimate reservoir evaporation.

Regression analysis determined that shared CRSP evaporation reported in Reclamation (2007a) was based on the average of reservoir contents at the beginning of the year and reservoir contents at the end of the year. Because year-end storage depends on the estimate of reservoir evaporation, an iterative approach is currently used for the annual simulation; referring to Appendix A, equations A1-A4 and A6 are iterated to convergence within 5 af to determine year-end storage. Carryover storage is then set (equation A5).

B.10 Curtailments

Curtailments are the amount by which UBCD must be reduced in any year to comply with the assumed composite flow obligation at Lee Ferry.

Appendix C. Water Balance Model Validation

The Water Balance Model was validated against Run 2, from Table A1 of 2007HD. Run 2 was used because it resulted in some shortages, which made comparison more straightforward. The validation run was conducted using the Lee Ferry natural flows from Reclamation (2007a), as these flows are different from those in Reclamation (2019) over the entire study period. The quantity of PPRs was set to zero, but this is a formality because the curtailments in the 2007 Hydrologic Determination did not reach the 2.8 bcm (2.267 maf) level described earlier. Because the water balance is simply an accounting, if the code is correct any error will be driven solely by the estimate of evaporation and will be very small.

The results of this simulation agreed very closely with the results for Run 2 published in Reclamation (2007a). Shortages, which are the variable of interest, and represent integration of conditions over extended periods, replicated values in Reclamation (2007a) within less than 0.001%.

This process did not test the validity of mechanisms in the Water Balance Model that were not exercised under the assumptions of Run 2. These mechanisms include protection of PPRs and conformance with the ten-year cumulative flow obligation at Lee Ferry. These mechanisms were validated by diagnosis and comparison to hand calculations.

References

Associated Press. 2008. Last-ditch effort to restart wells on Platte fails. Denver Post, May 3, 2008. https://www.denverpost.com/2008/05/03/last-ditch-effort-to-restart-wells-on-platte-fails/

Broadbent, C.D., Brookshire, D.S., Coursey, D. et al. 2017. Futures contracts in water leasing: an experimental analysis using basin characteristics of the Rio Grande, *NM. Environ Resource Econ* 68, 569–594 (2017). https://doi.org/10.1007/s10640-016-0032-4

Colorado Mesa University. 2021 Lessons Learned from Colorado Experiences with Interstate Compact Administration. https://www.coloradomesa.edu/water-center/compact-stories.html

Colorado River Basin Project Act. 82 stat 885. September 30, 1968. (See Nathanson, 1978.)

Colorado River Compact, November 24, 1922. *COLO. REV. STAT. § 37-61-101 (2012).* (See Nathanson, 1978.)

Colorado River Governance Initiative (CRGI). 2013. *The "Upper Basin Voluntary Demand Cap" as a means of mitigating legal uncertainty in the Colorado River Basin: modeling results*. April, 2013

Colorado River Governance Initiative (CRGI). 2010. *Rethinking the future of the Colorado River*, Draft Interim Report of the Colorado River Governance Initiative. December, 2010

Colorado Water Conservation Board (CWCB). 2012. *Colorado River water availability study*. Colorado Water Conservation Board. https://dnrweblink.state.co.us/cwcb/0/doc/158319/Electronic.aspx?searchid=78f0eafa-0b8f-4d8a-9ff3-faf67cc82f52.

Department of the Interior. 2007. Record of Decision Colorado River interim guidelines for lower basin shortages and the coordinated operations for Lake Powell and Lake Mead. U.S.

764 Department of Interior, December 2007.
765 https://www.usbr.gov/lc/region/programs/strategies/RecordofDecision.pdf
766 Getches, David. H. 1984. *Water law in a nutshell*. St. Paul MN, Thomson/Reuters.
767 Harding B.L., Wood A.W., Prairie J.R. 2012. The implications of climate change scenario
768 selection for future streamflow projection in the Upper Colorado River Basin. *Hydrol Earth*
769 *Syst Sci Discuss* 16: 3989–4007.
770 Harding, B.L. 2024.[Dataset] Reservoir Memory. https://doi.org/10.17605/OSF.IO/4M9DN
771 Hobbs, Gregory J. 1997. *Colorado water law: An historical overview*, 1 U. *Denv. Water L. Rev.* 1
772 Kenney, D. 2012a. Does the Upper Basin have a delivery obligation or an obligation not to
773 deplete the flow of the Colorado River at Lee Ferry? Western Water Policy Program,
774 Boulder Colorado, April 2012.
775 Kenney, D. 2012b. Respective obligations of the Upper and Lower Basins regarding the delivery
776 of water to Mexico: a review of key legal issues. Western Water Policy Program, Boulder
777 Colorado, March 2012.
778 Kenney, D. 2019. The case for the grand bargain. Colorado River District Annual Seminar.
779 Grand Junction, CO. September 18, 2019
780 Kuhn, E. 2012. Risk management strategies for the Upper Colorado River Basin. Colorado River
781 Water Conservation District. 2012. https://waterinfo.org/wp-
782 content/uploads/2019/01/Kuhn_on_Risk_Mgt_Strategies_of_the_UCRB.pdf
783 Kuhn, E. 2024. What Happens if there is no Agreement on Post-2026 Colorado River
784 Operations? https://www.inkstain.net/2024/04/what-happens-if-there-is-no-agreement-on-
785 post-2026-colorado-river-operations/
786 Kuhn, E. and Fleck, J. 2019. *Science be dammed: how ignoring inconvenient science drained the*
787 *Colorado River*. Tucson: University of Arizona Press. 288 pages.
788 Kuhn, E. and Jacobs, K. 2022. Robison, J.A. (2022). Alternative futures for the Colorado River system.
789 In *Cornerstone at the Confluence*: *Navigating the Colorado River Compact's Next Century*,
790 Robison, J.A. Ed., Tucson: University of Arizona Press, 2022.
791 https://doi.org/10.2307/j.ctv2z6qfp2.7
792 Lukas, Jeffery J., Personal communication, September 5, 2019.
793 MacDonnell, L., D.H. Getches, and W.C. Hugenberg Jr. 1995. The Law of the Colorado River:
794 Coping with Severe Sustained Drought. *Water Resources Bulletin*, 31:5(825-836), October.
795 MacDonnell, Lawrence. 2015. Prior Appropriation: A Reassessment, 18 U. *Denv. Water L. Rev.*
796 228 (2015)
797 MacDonnell, L. 2021a, Colorado River Basin. waters and water rights, Lexis-Nexus, CORB-1,
798 2021, Available at SSRN: https://ssrn.com/abstract=3780342
799 MacDonnell, L. 2021b. Sources of controversy in the law of the Colorado River: an Upper Basin
800 view. Available at SSRN: https://ssrn.com/abstract=3874212 or
801 http://dx.doi.org/10.2139/ssrn.3874212
802 Meko, D.M., C. A. Woodhouse, C. A. Baisan, T. Knight, J. J. Lukas, M. K. Hughes, & M. W.
803 Salzer. 2007. Medieval Drought in the Upper Colorado River Basin. *Geophysical Research*
804 *Letters* 34 (10). https://doi.org/10.1029/2007GL029988.
805 Mexican Water Treaty, 1944. *Treaty on utilization of the waters of the Colorado and Tijuana*
806 *Rivers and of the Rio Grande.* February 3, 1944, U.S. Government Printing Office, Treaty
807 Series 994. (See Nathanson, 1978.)
808 Miller, W. P., G.M. DeRosa, S. Gangopadhyay and J.B. Valdes. 2013. Predicting regime shifts in
809 flow of the Gunnison River under changing climate conditions. *Water Resources Research*,
810 v 49, 2966–2974, doi:10.1002/wrcr.20215.
811 MWH, 2012. *Colorado River Water Bank Feasibility Study – Phase 1, Final Draft Report.*
812 MWH Americas, Inc., Denver, Colorado. June, 2012.

Nathanson, M.N. 1978. *Updating the Hoover Dam Documents*. United States Department of the Interior Bureau of Reclamation. U.S. Government Printing Office, Denver, Colorado, 1980.

Reclamation, 1970. *Criteria for coordinated long-range operation of Colorado River reservoirs*. June, 1970.

Reclamation. 2007a. *Hydrologic determination: water availability from Navajo Reservoir and the Upper Colorado River Basin for use in New Mexico*. April, 2007.

Reclamation. 2007b. *Final Environmental Impact Statement for the Colorado River interim guidelines for Lower Basin shortages and coordinated operations for Lake Powell and Lake Mead;* Appendix N– Analysis of Hydrologic Variability Sensitivity. https://www.usbr.gov/lc/region/programs/strategies/FEIS/index.html. U.S. Department of the Interior, Bureau of Reclamation, Upper and Lower Colorado Regions. October, 2007.

Reclamation. 2012. *Colorado River Basin water supply and demand study, study report*. U.S. Department of Interior, Bureau of Reclamation. December, 2012.

Reclamation. 2019. *Natural flow and salt records*. https://www.usbr.gov/lc/region/g4000/NaturalFlow/NaturalFlows1906-2017_3.18.2019.xlsx Last accessed September 12, 2019. U.S. Department of the Interior, Bureau of Reclamation Lower Colorado Region, Boulder Canyon Operations Office.

Rice, J. L., C. A. Woodhouse, and J. J. Lukas. 2009. Science and decision making: water management and tree-ring data in the western united states. *JAWRA Journal of the American Water Resources Association* 45 (5): 1248–59. https://doi.org/10.1111/j.1752-1688.2009.00358.x.

Robison, J. A. & Kenney, D. S. 2012. Equity and the Colorado River Compact. *Environmental Law* 42, 1157–1209.

Treeflow.info. https://www.treeflow.info/ Data retrieved on November 20, 2018; last access on September 6, 2019.

Udall, B., and J. T. Overpeck. 2017: The twenty-first century Colorado River hot drought and implications for the future. *Water Resour. Res.*, 53, 2404–2418, doi:https://doi.org/10.1002/2016WR019638.

Upper Colorado River Basin Compact. 1948. October 11, 1948. (See Nathanson, 1978.)

Verburg, K. O. 2010. *The Colorado River documents 2008*. U.S. Department of the Interior, Bureau of Reclamation, Lower Colorado River Region. September 2010

Weisheit, J. 2019. *The Hoover Dam documents*. http://www.onthecolorado.com/articles.cfm?mode=detail&id=1292710182151 Accessed September 22, 2019.

Wheeler, K. 2019. The "Upper Basin Demand Cap": a new interbasin grand bargain? in *Charting a better course for the Colorado River: identifying the data and concepts to shape the interim guidelines renegotiation*. Getches-Wilkinson Center for Natural Resources, Energy, and the Environment. June, 2019.

Wigington, R. 2019. Personal communication. September 14, 2019.

Wilbur, R. L., Ely, N. 1933. *The Hoover Dam power and water contracts and related data*. United States Department of the Interior, United States Government Printing Office. Washington, D.C. 1933.

Wilbur, R. L., Ely, N. 1948. *The Hoover Dam documents*. 80th Congress 2nd Session, House Document No. 717. United States Government Printing Office. Washington, D.C. 1948.